Math 125 Class notes
Spring 2026

To accompany
Discrete Mathematics with Applications
by *Epp*

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1.1: Variables

Definition.
A **variable** is a placeholder for something which may or may not be unknown.

Example. Is there a number with the following property: doubling it and adding 3 gives the same result as squaring it?

$\rightarrow$ Is there a number x with the property that $2x + 3 = x^2$?
$\rightarrow$ Is there a number $\square$ with the property that $2 \cdot \square + 3 = \square^2$?

Example. No matter what number might be chosen, if it is greater than 2, then its square is greater than 4.

$\rightarrow$ No matter what number n might be chosen, if n is greater than 2, then n^2 is greater than 4.

Example. Use variables to rewrite the following sentences:

Are there numbers with the property that the sum of their squares equals the square of their sum?

Given any real number, its square is nonnegative.

Definition.

- A **universal statement** says that a certain property is true for all elements in a set.
- A **conditional statement** says that if one thing is true, then some other thing also has to be true.
- Given a property that may or may not be true, an **existential statement** says that there is at least one thing for which the property is true.

Definition.
A **universal conditional statement** is both universal and conditional:

For every animal a, if a is a dog, then a is a mammal.

Conditional statements can be rewritten in ways that make them appear more to be purely universal or purely conditional:

If a is a dog, then a is a mammal.
All dogs are mammals

Example. Rewrite the following universal condition statement:

For every real number x, if x is nonzero then x^2 is positive.

If a real number is nonzero, then its square _____.

For every nonzero real number x, _____.

If x _____, then _____.

The square of any nonzero real number is _____.

All nonzero real numbers have _____.

Definition.
A **universal existence statement** is a statement that is universal because its first part says that a certain property is true for all objects of a given type, and it is existential because its second part asserts the existence of something:

Every real number has an additive inverse.

In the above example, note that the particular additive inverse depends on the given real number:

For every real number r, there is an additive inverse for r.

Example. Rewrite the following universal existence statement:

Every pot has a lid

All pots _____.

For ever pot P, there is _____.

For every pot P, there is a lid L such that _____.

Definition.
An **existential universal statement** is a statement that is existential because its first part asserts that a certain object exists and is universal because its second part says that the object satisfies a certain property for all things of a certain kind:

There is a positive integer that is less than or equal to every positive integer.

The number one satisfies the above statement, which can also be rewritten:

There is a positive integer m that is less than or equal to every positive integer.

Example. Rewrite the following existence universal statement:

There is a person in my class who is at least as old as every person in my class.

Some _____ is at least as old as _____.

There is a person p in my class such that p is _____.

There is a person p in my class with the property that for every person q in my class, p is _____.

1.2: The Language of Sets

Definition.

- A **set** is a collection of objects.
- If S is a set, then we use
  - $x \in S$ to denote that the element x is in the set S.
  - $x \notin S$ to denote that the element x is *not* in the set S.
- The **set-roster notation** is used to denote all elements in a set between braces:

$$S = \{1, 2, \ldots, 100\}$$

  Here, we see that $67 \in S$, but $1337 \notin S$.
- The **axiom of extension** says that a set is completely determined by what its elements are – not the order in which they are listed.

Example.

Let $A = \{1, 2, 3\}$, $B = \{3, 1, 2\}$, and $C = \{1, 1, 2, 3, 3, 3\}$. What are the elements of A, B, and C? How are A, B, and C related?

Is $\{0\} = 0$?

How many elements are in the set $\{1, \{1\}\}$?

For each nonnegative integer n, let $U_n = \{n, -n\}$. Find U_1, U_2, and U_0.

Certain sets of numbers are so frequently referred to that they are given special names and symbols:

N or $\mathbb{N}$	The set of all **natural numbers**
Z or $\mathbb{Z}$	The set of all **integers**
Q or $\mathbb{Q}$	The set of all **rational numbers**, or quotient of integers
R or $\mathbb{R}$	The set of all **real numbers**

Note: We may additionally use superscripts to indicate further properties of these sets:

$\mathbb{Z}^+$ or $\mathbb{Z}^{>0}$	The set of *positive* integers
$\mathbb{Q}^-$ or $\mathbb{Q}^{<0}$	The set of *negative* rational numbers
$\mathbb{R}^{nonneg}$ or $\mathbb{R}^{\geq 0}$	The set of *nonnegative* real numbers

Note: Different sources denote the natural numbers $\mathbb{N}$ as $\mathbb{Z}^+$ *or* $\mathbb{Z}^{\geq 0}$.

Definition. (Set-Builder Notation)
Let S be a set and let $P(x)$ be a property that elements of S may or may not satisfy. We may define a new set to be **the set of all elements x in S such that $P(x)$ is true**. We denote this set as follows:

$$\{x \in S | P(x)\}$$

the set of all such that

Example. Describe each of the following sets:

$\{x \in \mathbb{R} \mid -2 < x < 5\}$

$\{x \in \mathbb{Z} \mid -2 < x < 5\}$

$\{x \in \mathbb{Z}^+ \mid -2 < x < 5\}$

Definition.
If A and B are sets, then A is called a **subset** of B, written $A \subseteq B$, if, and only if, every element of A is also an element of B:

$$A \subseteq B \text{ means that for every element } x, \text{ if } x \in A, \text{ then } x \in B.$$
$$A \nsubseteq B \text{ means that there is at least one element } x, \text{ such that } x \in A \text{ and } x \notin B.$$

A is a **proper subset** of B if, and only if, every element of A is in B, but there is at least one element of B that is not in A:

$$A \subsetneq B \text{ means that for every element } x, \text{ if } x \in A, \text{ then } x \in B,$$
$$\text{and there exists } x \in B \text{ such that } x \notin A.$$

Example. Let $A = \mathbb{Z}^+$, $B = \{n \in \mathbb{Z} \mid 0 \leq n \leq 100\}$, and $C = \{100, 200, 300, 400, 500\}$. Evaluate the truth and falsity of each of the following statements.

$B \subseteq A$

C is a proper subset of A

C and B have at least one element in common

$C \subseteq B$

$C \subseteq C$

Example. Determine which of the following statements are true:

$2 \in \{1, 2, 3\}$

$\{2\} \in \{1, 2, 3\}$

$2 \subseteq \{1, 2, 3\}$

$\{2\} \subseteq \{1, 2, 3\}$

$\{2\} \subseteq \{\{1\}, \{2\}\}$

$\{2\} \in \{\{1\}, \{2\}\}$

Definition.
Given elements a and b, the symbol (a, b) denotes the **ordered pair** consisting of a and b together with the specification that a is the first element of the pair, and b is the second element. Two ordered pairs (a, b) and (c, d) are equal if, and only if, $a = c$ and $b = d$:

$$(a, b) = (c, d) \text{ means that } a = c \text{ and } c = d.$$

Example.

Is $(1, 2) = (2, 1)$?

Is $\left(3, \frac{5}{10}\right) = \left(\sqrt{9}, \frac{1}{2}\right)$?

Definition.
Let $n \in \mathbb{N}$ and let $x_1, x_2, \ldots, x_n$ be (not necessarily distinct) elements. The **ordered n-tuple**, $(x_1, x_2, \ldots, x_n)$, consists of $x_1, x_2, \ldots x_n$ together with the ordering: first x_1, then x_2, and so forth up to x_n. and ordered 2-tuple is called an **ordered pair**, and ordered 3-tuple is called an **ordered triple**.
Two ordered n-tuples $(x_1, x_2, \ldots, x_n)$ and $(y_1, y_2, \ldots, y_n)$ are **equal** if, and only if, $x_1 = y_1$, $x_2 = y_2$, $\ldots$, and $x_n = y_n$:

$$(x_1, x_2, \ldots, x_n) = (y_1, y_2, \ldots, y_n) \iff x_1 = y_1, x_2 = y_2, \ldots, x_n = y_n.$$

Example.

Is $(1, 2, 3, 4) = (1, 2, 4, 3)$?

Is $\left(3, (-2)^2, \frac{1}{2}\right) = \left(\sqrt{9}, 4, \frac{3}{6}\right)$?

Definition.
Given sets $A_1, A_2, \ldots, A_n$, the **Cartesian product** of $A_1, A_2, \ldots, A_n$, denoted

$$A_1 \times A_2 \times \cdots \times A_n$$

is the set of all ordered n-tuples $(a_1, a_2, \ldots, a_n)$ where $a_1 \in A_1$, $a_2 \in A_2$, $\ldots$, $a_n \in A_n$:

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in \mathcal{A}_n\}$$

Example. Let $A = \{x, y\}$, $B = \{1, 2, 3\}$, and $C = \{a, b\}$. Find the following:

$A \times B$

$B \times A$

$A \times A$

How many elements are in $A \times B$, $B \times A$, and $A \times A$?

$(A \times B) \times C$

$A \times B \times C$

Describe $\mathbb{R} \times \mathbb{R}$

Definition.
Let $n \in \mathbb{N}$. Given a finite set A, a **string of length** n **over** A is an ordered n-tuple of elements of A written without parentheses or commas. The elements of A are called the **characters** of the string. The **null string** over A is defined to be the "string" with no characters, often denoted λ, and is said to have length 0. If $A = \{0, 1\}$, then a string over A is called a **bit string**.

Example. Let $A = \{a, b\}$. List all strings of length 3 over A with at least two characters that are the same.

1.3: The Language of Relations and Functions

Definition.
Let A and B be sets. A **relation** R **from** A **to** B is a subset of $A \times B$. Given an ordered pair (x, y), x **is related to** y **by** R, written $x\,R\,y$, if, and only if, (x, y) is in R. The set A is called the **domain** of R and the set B is called its **co-domain**.
The notation for a relation R may be written symbolically as follows:

$$x\,R\,y \text{ means that } (x, y) \in R.$$

The notation $x \not R\, y$ means that x is not related to y by R:

$$x \not R\, y \text{ means that } (x, y) \notin R.$$

Example. Let $A = \{1, 2\}$ and $B = \{1, 2, 3\}$ and define a relation R from A to B as follows; Given any $(x, y) \in A \times B$,

$$(x, y) \in R \text{ means that } \frac{x - y}{2} \text{ is an integer.}$$

State explicitly which ordered pairs are in $A \times B$ and which are in R

Is $1\,R\,3$? Is $2\,R\,3$? Is $2\,R\,2$?

What are the domain and co-domain of R?

Example. Define a relation C from $\mathbb{R}$ to $\mathbb{R}$ as follows: For any $(x, y) \in \mathbb{R} \times \mathbb{R}$,

$$(x, y) \in C \text{ means that } x^2 + y^2 = 1.$$

Is $(1, 0) \in C$?

Is $(0, 0) \in C$?

Is $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \in C$?

Is $-2\, C\, 0$?

Is $0\, C\, (-1)$?

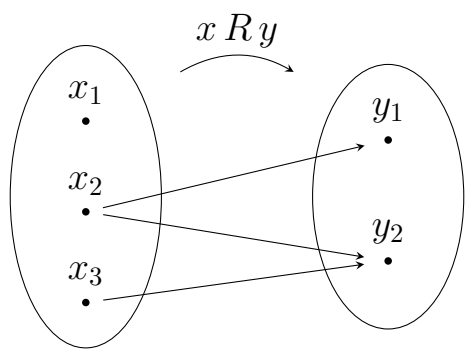
Is $1\, C\, 1$?

What are the domain and co-domain of C?

Draw a graph for C by plotting the points of C in the Cartesian plane.

Definition.
Suppose R is a relation from set A to a set B. The **arrow diagram for** R is obtained as follows:

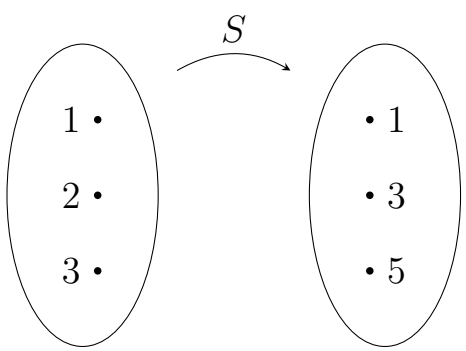
1. Represent the elements of A as points in one region and the elements of B as points in another region.
2. For each x in A and y in B, draw an arrow from x to y if, and only if, x is related to y by R.

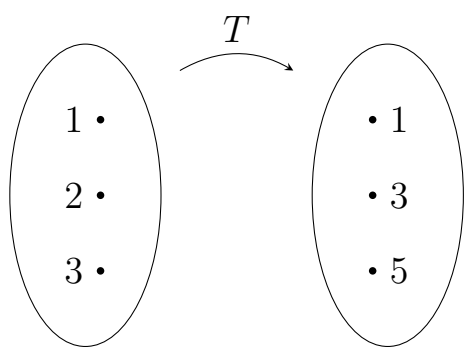


Example. Let $A = \{1, 2, 3\}$ and $B = \{1, 3, 5\}$ and define relations S and T from A to B as follows: For every $(x, y) \in A \times B$,

$$(x, y) \in S \text{ means that } x < y$$
$$T = \{(2, 1), (2, 5)\}.$$

Draw arrow diagrams for S and T





Definition.
A **function** F **from a set** A **to a set** B is a relation with domain A and co-domain B that satisfies the following two properties:

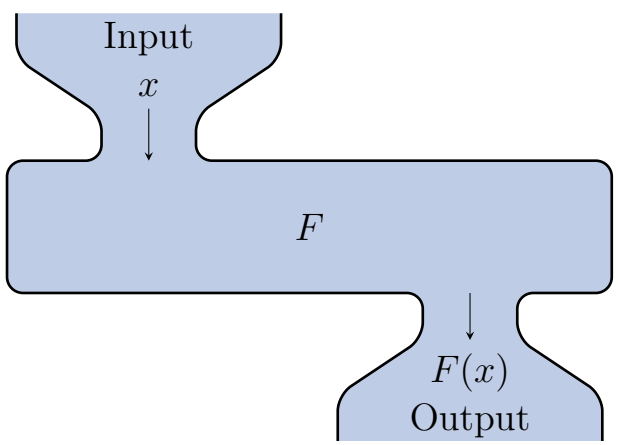
1. For every element x in A, there is an element y in B such that $(x, y) \in F$.
2. For all elements x in A and y and z in B,

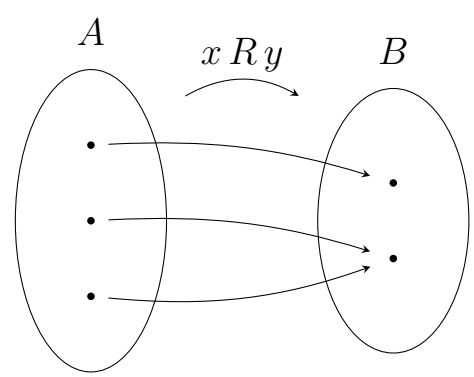
$$\text{if } (x, y) \in F \text{ and } (x, z) \in F, \text{ then } y = z.$$

Note: A relation from A to B is a function if, and only if,

1. Every element of A is the first element of an ordered pair of F
2. No two distinct ordered pairs in F have the same first element.

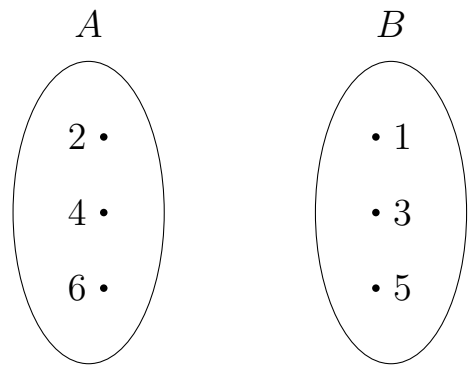
Note: If A and B are sets and F is a function from A to B, then given any element x in A, the unique element in B that is related to x by F is denoted $F(x)$, which is read "F **of** x".



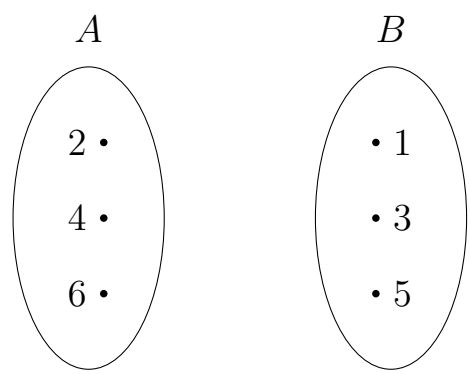


Example. Let $A = \{2, 4, 6\}$ and $B = \{1, 3, 5\}$. Which of the relations R, S, and T defined below are functions from A to B?

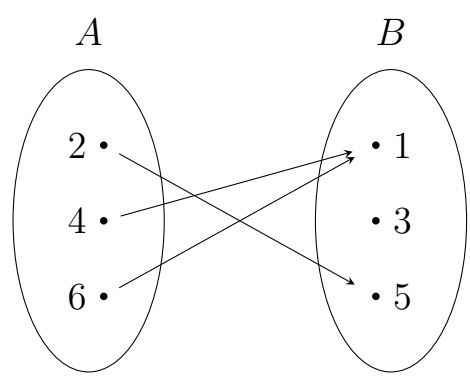
$R = \{(2, 5), (4, 1), (4, 3), (6, 5)\}$



For every $(x, y) \in A \times B$, $(x, y) \in S$ means that $y = x + 1$.



T is defined by the arrow diagram



2.1: Logical Form and Logical Equivalence

Definition.
A **statement** (or **proposition**) is a sentence that is true or false, but not both.

Example. Determine which of the following are statements:

$2 + 2 = 4$

$2 + 2 = 5$

$x^2 + 2 = 11$

Today is Saturday.

She is a computer science major.

Jane is a computer science major.

Definition. (Compound Statements)
Let p and q be statement variables.

- The **negation** of p is "not p", and is denoted as $\sim p$ (or $\neg p$)
- The **conjunction** of p and q is "p and q", and is denoted at $p \wedge q$
- The **disjunction** of p and q is "p or q", and is denoted $p \vee q$.
- The **exclusive or** of p and q is "p x-or q", and is denoted $p \oplus q$ (or p XOR q)

The **order of operations** specifies that $\sim$ is performed first.

Example. Consider the following statements:

$$\begin{aligned} p &: \text{ It is raining.} \\ q &: \text{ It is sunny.} \\ r &: \text{ It is cloudy.} \end{aligned}$$

Rewrite the following compound statements in words:

$\sim p$

$p \vee q$

$q \wedge r$

$q \wedge \sim r$

$p \wedge (q \vee r)$

$p \oplus q$

Definition.
A **statement form** (or **propositional form**) is an expression made up of statement variables (e.g., p, q, and r), and logical connectives (e.g. $\sim$, $\wedge$, $\vee$, and $\oplus$).
The **truth table** for a given statement form displays the truth values that correspond to all possible combinations of truth values for its component statement variables.

Example. Let p and q be statement variables. Fill out the following truth tables:

p	$\sim p$
T	
F	

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$
T	T			
T	F			
F	T			
F	F			

p	q	$p \vee q$	$p \wedge q$	$\sim (p \wedge q)$	$(p \vee q) \wedge \sim (p \wedge q)$
T	T				
T	F				
F	T				
F	F				

Example. Construct a truth table for the statement form $(p \wedge q) \vee \sim r$.

Definition.
Two *statement forms* are called **logically equivalent** if, and only if, they have identical true values for each possible substitution of statements for their statement variables. The logical equivalence of statement forms P and Q is denoted $P \equiv Q$.

Example. Use truth tables to test if the following statement forms are equivalent:

p and $\sim(\sim p)$

$\sim(p \wedge q)$ and $\sim p \wedge \sim q$

Definition. (De Morgan's Laws)
The negation of an *and* statement is logically equivalent to the *or* statement in which each component is negated.
The negation of an *or* statement is logically equivalent to the *and* statement in which each component is negated.

Example. Use truth tables to show that the following statement forms are equivalent:

$\sim (p \wedge q)$ and $\sim p \vee \sim q$

$\sim (p \vee q)$ and $\sim p \wedge \sim q$

Example. Using De Morgan's law to write the negation of the following statements:

Jim is at least 6 feet tall and weighs at least 200 pounds.

The bus was late or Tom's watch was slow.

$-1 < x \leq 4$

Definition.
A **tautology** is a statement form that is always true.
A **contradiction** is a statement form that is always false.

Example. Complete the truth tables for $p \wedge \sim p$ and $p \vee \sim p$

Example. Let $\mathbf{t}$ be a tautology, and $\mathbf{c}$ be a contradiction. Show that $p \wedge \mathbf{t} \equiv p$ and $p \wedge \mathbf{c} \equiv \mathbf{c}$

Theorem 2.1.1 Logical Equivalences (p 49)
Given any statement variables p, q, and r, a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold:

1. Commutative laws:
   $p \wedge q \equiv q \wedge p$ $\qquad$ $p \vee q \equiv q \vee p$
2. Associative laws:
   $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $\qquad$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$
3. Distributive laws:
   $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $\qquad$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
4. Identity laws:
   $p \wedge \mathbf{t} \equiv p$ $\qquad$ $p \vee \mathbf{c} \equiv p$
5. Negation laws:
   $p \vee \sim p \equiv \mathbf{t}$ $\qquad$ $p \wedge \sim p \equiv \mathbf{c}$
6. Double negative law:
   $\sim (\sim p) \equiv p$
7. Idempotent laws:
   $p \wedge p \equiv p$ $\qquad$ $p \vee p \equiv p$
8. Universal bound laws:
   $p \vee \mathbf{t} \equiv \mathbf{t}$ $\qquad$ $p \wedge \mathbf{c} \equiv \mathbf{c}$
9. De Morgan's laws:
   $\sim (p \wedge q) \equiv \sim p \vee \sim q$ $\qquad$ $\sim (p \vee q) \equiv \sim p \wedge \sim q$
10. Absorption laws:
    $p \wedge (p \vee q) \equiv p$ $\qquad$ $p \wedge (p \vee q) \equiv p$
11. Negations of $\mathbf{t}$ and $\mathbf{c}$:
    $\sim \mathbf{t} \equiv \mathbf{c}$ $\qquad$ $\sim \mathbf{c} \equiv \mathbf{t}$

2.2: Conditional Statements

Definition.
If p and q are statement variables, the **conditional** of q by p is "If p then q", or "p implies q" and is denoted by $p \to q$. It is false when p is true and q is false; otherwise it is true. We call p the **hypothesis** (or **antecedent**) of the conditional and q the **conclusion** (or **consequent**).
A conditional statement that is always true because the hypothesis is false is called **vacuously true**.

$$\text{If } \underbrace{4,686 \text{ is divisible by } 6,}_{\text{hypothesis}} \text{ then } \underbrace{4,686 \text{ is divisible by } 3}_{\text{conclusion}}$$

Example. Consider the following statement:

If Lander is open, then we will have class.

Create the truth table for $p \to q$

p	q	$p \to q$
T	T	
T	F	
F	T	
F	F	

Note: The **order of operations** states that $\rightarrow$ is performed last

Example. Create the truth table for $p \vee \sim q \rightarrow \sim p$.

p	q	$\sim q$	$p \vee \sim q$	$\sim p$	$p \vee \sim q \rightarrow \sim p$
T	T				
T	F				
F	T				
F	F				

Example. Use a truth table to show that $p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

p	q	r	$p \vee q$	$p \vee q \rightarrow r$	$p \rightarrow r$	$q \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
T	T	T					
T	T	F					
T	F	T					
T	F	F					
F	T	T					
F	T	F					
F	F	T					
F	F	F					

Definition.
The **negation** of "if p then q" is logically equivalent to "p and not q":

$$\sim (p \to q) \equiv p \wedge \sim q$$

Example. Write negations for each of the following statements:

If my car is in the repair shop, then I cannot get to class.

If Sara lives in Athens, then she lives in Greece.

Definition.
The **contrapositive** of a conditional statement of the form "If p then q" is

$$\text{If } \sim q \text{ then } \sim p: \quad \sim q \to \sim p$$

A conditional statement is logically equivalent to its contrapositive.

Example. Write each of the following statements in its equivalent contrapositive form:

If Howard can swim across the lake, then Howard can swim to the island.

If today is Easter, then tomorrow is Monday.

Definition.
Suppose a conditional statement of the form "If p then q" is given.

- The **converse** is "If q then p": $q \to p$
- The **inverse** is "If $\sim p$ then $\sim q$": $\sim p \to \sim q$

Example. Write the converse and inverse of each of the following statements:

If Howard can swim across the lake, then Howard can swim to the island.

Converse:

Inverse:

If today is Easter, then tomorrow is Monday.

Converse:

Inverse:

Note:

1. A conditional statement and its converse are *not* logically equivalent.
2. A conditional statement and its inverse are *not* logically equivalent.
3. The converse and the inverse of a conditional statement are logically equivalent to each other.

Definition.
If p and q are statements, p **only if** q means "if not q then not p":

$$\sim q \rightarrow \sim p \equiv p \rightarrow q$$

Example. Rewrite the following statement in if-then form in two ways, one of which is the contrapositive of the other:

> John will break the world's record for the mile run only if he runs the mile in under four minutes.

$\sim q \rightarrow \sim p$

$p \rightarrow q$

Note:

1. "p only if q" does *not* mean p if q
2. It is possible for "p only if q" to be true at the same time that "p if q" is false.

   e.g.: If John runs a mile in under four minutes, he still might not be fast enough to break the record.

Definition.
Given statement variables p and q, the **biconditional of p and q** is "p if, and only if, q" and is denoted $p \leftrightarrow q$. It is true if both p and q have the same truth values and is false otherwise. The words *if and only if* are sometimes abbreviated **iff**.

Note: The **order of operations** states that $\leftrightarrow$ is coequal with $\rightarrow$

Example. Create the truth table for $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	
T	F	
F	T	
F	F	

Order of Operations for Logical Operators

$\sim$	Evaluate negations first
$\wedge, \vee$	Evaluate $\wedge$ and $\vee$ second. When both present, parentheses may be needed.
$\rightarrow, \leftrightarrow$	Evaluate $\rightarrow$ and $\leftrightarrow$ third. When both present, parentheses may be needed.

Definition.
If r and s are statements:

1. r is a **sufficient condition** for s means "if r then s". $r \to s$

2. r is a **necessary condition** for s means "if not r then not s". $\sim r \to \sim s$

By property of the contrapositive:

3. r is a *necessary and sufficient condition* for s means "r if, and only if s."
$r \leftrightarrow s$

Example. Rewrite the following statement in the form "If A then B":

Having two 45° angles is a sufficient condition for this triangle to be a right triangle.

Example. Use the contrapositive to rewrite the following statement in two ways:

George's attaining age 35 is a necessary condition for his being president of the United States.

2.5: Application: Number Systems and Circuits for Addition

Recall our how we write numbers in base 10:

$$\begin{aligned} 5,049 =& 5\cdot 1000 + 0\cdot 100 + 4\cdot 10 + 9\cdot 1 \\ =& 5\cdot 10^3 + 0\cdot 10^2 + 4\cdot 10^1 + 9\cdot 10^0 \end{aligned}$$

Definition.
Any integer $b > 1$ can be used as a base for a numbering system. A numbering system of base b has the digits $0, 1, \ldots, b-1$.
A **base** 2 **notation** or **binary notation**, uses the digits $0, 1$. In binary, every integer is represented as sum of products of the form

$$d \cdot 2^n$$

where $n \in \mathbb{Z}$ and $d \in \{0, 1\}$.

Example. Below is the binary representation for the integers 1 to 9:

$$\begin{array}{lcrcrcrcr}
1_{10} & = & & & & & 1\cdot 2^0 & = & 1_2 \\
2_{10} & = & & & 1\cdot 2^1 & + & 0\cdot 2^0 & = & 10_2 \\
3_{10} & = & & & 1\cdot 2^1 & + & 1\cdot 2^0 & = & 11_2 \\
4_{10} & = & & 1\cdot 2^2 & + & 0\cdot 2^1 & + & 0\cdot 2^0 & = & 100_2 \\
5_{10} & = & & 1\cdot 2^2 & + & 0\cdot 2^1 & + & 1\cdot 2^0 & = & 101_2 \\
6_{10} & = & & 1\cdot 2^2 & + & 1\cdot 2^1 & + & 0\cdot 2^0 & = & 110_2 \\
7_{10} & = & & 1\cdot 2^2 & + & 1\cdot 2^1 & + & 1\cdot 2^0 & = & 111_2 \\
8_{10} & = & 1\cdot 2^3 & + & 0\cdot 2^2 & + & 0\cdot 2^1 & + & 0\cdot 2^0 & = & 1000_2 \\
9_{10} & = & 1\cdot 2^3 & + & 0\cdot 2^2 & + & 0\cdot 2^1 & + & 1\cdot 2^0 & = & 1001_2
\end{array}$$

Converting binary $\rightarrow$ decimal:
To convert from binary to decimal, multiply each digit by its corresponding power of 2 and sum the results.

Example. Represent the following in decimal notation (base-10):

110_2

1011_2

11110_2

101011_2

Converting decimal $\to$ binary:
To convert from decimal to binary, we repeated divide by 2, and record the remainders.

Example.

$$\begin{aligned} 27_{10} &= 16 + 8 + 2 + 1 \\ &= 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \\ &= 11011_2 \end{aligned}$$

Example. Represent the following in binary notation:

243_{10}

587_{10}

990_{10}

531_{10}

Binary arithmetic:
In binary arithmetic, 10_2 behaves similarly to 10 in decimal arithmetic.

Example. Add 1101_2 and 111_2 using binary notation.

Example. Subtract 1011_2 from 11000_2 using binary notation.

Definition.
The 8-bit two's complement for an integer a between -128 and 127 is the 8-bit binary representation for

$$\begin{cases} a, & \text{if } a \geq 0 \\ 2^8 - |a|, & \text{if } a < 0. \end{cases}$$

Two's complement allows maximum representation for 2^8 integers with 8 binary digits.

Example. Below are a few integers represented in binary using 8-bit two's complement:

$$-128 \to 2^8 - |-128| = 128_{10} = 10000000_2 \qquad 0 \to 0_{10} = 00000000_2$$

$$-127 \to 2^8 - |-127| = 129_{10} = 10000001_2 \qquad 1 \to 1_{10} = 00000001_2$$

$$\vdots \qquad 2 \to 2_{10} = 00000010_2$$

$$-2 \to 2^8 - |-2| = 254_{10} = 10000000_2 \qquad \vdots$$

$$-1 \to 2^8 - |-1| = 255_{10} = 11111111_2 \qquad 127 \to 127_{10} = 01111111_2$$

Example. Find the 8-bit two's complement for the following:

-46 $\qquad$ 42

120 $\qquad$ -82

Two's complement of a negative integer:
To find the decimal representation of the negative integer with a given 8-bit two's complement:

- Flip the bits
- Add 1
- Convert to base-10 and swap the sign

Example. Find the decimal representation of the integers with the following 8-bit two's complement:

11100101_2

11000000_2

Addition and Subtraction with Integers in Two's Complement Form: When performing binary addition on integers written in Two's Complement form, we discard any "carry" bit.

Example. Perform binary addition using the Two's Complement form of the following:

83 and -55

-87 and -46

Definition.
Hexadecimal notation uses a **base** 16 **notation**. In hexadecimal, every integer is represented as sum of products of the form

$$d \cdot 16^n$$

where $n \in \mathbb{Z}$ and $d \in \{0, 1, \ldots, 9, A, B, C, D, E, F\}$.

Decimal	Hexadecimal	4-Bit Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Example. Convert $3CF_{16}$ to decimal notation.

Example. Convert $B09F_{16}$ to binary notation.

Example. Convert 100110110101001_2 to hexadecimal notation.

3.1: Predicates and Quantified Statements I

Definition.
A **predicate** is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables.
The **domain** of a predicate variable is the set of all values that may be substituted in place of the variable.

Example. Let $P(x)$ be the predicate "$x^2 > x$" with domain the set $\mathbb{R}$. Write $P(2)$, $P\left(\frac{1}{2}\right)$, and $P\left(-\frac{1}{2}\right)$, and indicate which of these statements are true and which are false.

Definition.
If $P(x)$ is a predicate and x has domain D, the **truth set** of $P(x)$ is the set of all elements of D that make $P(x)$ true when they are substituted for x. The truth set of $P(x)$ is denoted

$$\{x \in D \mid P(x)\}$$

Example. Let $Q(n)$ be the predicate "n is a factor of 8". Find the truth set of $Q(n)$ if

the domain of n is $\mathbb{Z}^+$

the domain of n is $\mathbb{Z}$

Definition.
Let $Q(x)$ be a predicate and D the domain of x.

- **Quantifiers** are words that refer to quantities such as "some" or "all" and tell for how many elements a given predicate is true.
- The **universal quantifier** is represented by the symbol "$\forall$".
- A **universal statement** is a statement of the form "$\forall x \in D$, $Q(x)$".
  - It is defined to be true if, and only if, $Q(x)$ is true for *each* individual x in D.
  - It is defined to be false if, and only if, $Q(x)$ is false for *at least one* x in D.
- A value for x for which $Q(x)$ is false is called a **counterexample** to the universal statement.

Example. Let $D = \{1, 2, 3, 4, 5\}$, and consider the statement

$$\forall x \in D, x^2 \geq x.$$

Write one way to read this statement out loud, and show that it is true.

The above example uses the **method of exhaustion**.

Example. Consider the statement

$$\forall x \in \mathbb{R}, x^2 \geq x.$$

Find a counter example to show that this statement is false.

Definition.
Let $Q(x)$ be a predicate and D the domain of x.

- The **existential quantifier** is represented by the symbol "$\exists$".
- An **existential statement** is a statement of the form "$\exists x \in D$ such that $Q(x)$".
  - It is defined to be true if, and only if, $Q(x)$ is true for *at least one* x in D.
  - It is false if, and only if, $Q(x)$ is false *for all* x in D.

Example. Consider the statement

$$\exists m \in \mathbb{Z}^+ \text{ such that } m^2 = m.$$

Write one way to read this statement out loud, and show that it is true.

Example. Let $E = \{5, 6, 7, 8\}$ and consider the statement

$$\exists m \in E \text{ such that } m^2 = m.$$

Show that this statement is false.

Example. Rewrite the following statements formally using quantifiers and variables:

All triangles have three sides.

No dogs have wings.

Some programs are structured.

Definition.
A **universal conditional statement** is of the form:

$$\forall x, \text{ if } P(x) \text{ then } Q(x).$$

Example. Rewrite each of the following statements in the form

$$\forall _____, \text{ if } _____ \text{ then } _____$$

If a real number is an integer, then it is a rational number.

All bytes have eight bits.

No fire trucks are green.

3.2: Predicates and Quantified Statements II

Definition.

- The negation of a statement of the form

$$\forall x \text{ in } D,\ Q(x)$$

is logically equivalent to a statement of the form

$$\exists x \text{ in } D \text{ such that } \sim Q(x).$$

$$\sim (\forall x \in D,\ Q(x)) \equiv \exists x \in D \text{ such that } \sim Q(x).$$

- The negation of a statement of the form

$$\exists x \text{ in } D \text{ such that } Q(x)$$

is logically equivalent to a statement of the form

$$\forall x \text{ in } D,\ \sim Q(x).$$

$$\sim (\exists x \in D \text{ such that } Q(x)) \equiv \forall x \in D,\ \sim Q(x)$$

Example. Negate the following statements:

$\forall$ primes p, p is odd

$\exists$ a triangle T such that the sum of the angles of T equals $200°$

Example. Rewrite the following statements formally, then write the formal and informal negations.

No politicians are honest

The number $1,357$ is not divisible by any integer between 1 and 37.

Example. Write informal negations for the following statements:

All computer programs are finite.

Some computer hackers are over 40.

Negation of a Universal Conditional Statement

$$\sim (\forall x, \text{ if } P(x) \text{ then } Q(x)) \equiv \exists x \text{ such that } P(x) \text{ and } \sim Q(x)$$

Definition.
A statement of the form

$$\forall x \text{ in } D, \text{ if } P(x) \text{ then } Q(x)$$

is called **vacuously true** or **true by default** if, and only if, $P(x)$ is false for every x in D.

Example. The following statement is vacuously true since it's negation is false:

All kangaroos enrolled in my class are passing.

Definition.
Consider a statement of the form $\forall x \in D$, if $P(x)$ then $Q(x)$.

1. Its **contrapositive** is the statement $\forall x \in D$, if $\sim Q(x)$ then $\sim P(x)$.
2. Its **converse** is the statement $\forall x \in D$, if $Q(x)$ then $P(x)$.
3. Its **inverse** is the statement $\forall x \in D$, if $\sim P(x)$ then $\sim Q(x)$.

Example. Write a formal and informal contrapositive, converse, and inverse for the following statement:

If a real number is greater than 2, then its square is greater than 4.

Definition.

- "$\forall x, r(x)$ is a **sufficient condition** for $s(x)$" $\rightarrow$ "$\forall x$, if $r(x)$ then $s(x)$"
- "$\forall x, r(x)$ is a **necessary condition** for $s(x)$" $\rightarrow$ "$\forall x$, if $\sim r(x)$ then $\sim s(x)$"
  $\rightarrow$ "$\forall x$, if $s(x)$ then $r(x)$"
- "$\forall x, r(x)$ **only if** $s(x)$" $\rightarrow$ "$\forall x$, if $\sim s(x)$, then $\sim r(x)$"
  $\rightarrow$ "$\forall x$, if $r(x)$ then $s(x)$"

Example. Rewrite each of the following as a universal conditional statement, quantified either explicitly or implicitly. Do not use the word *necessary* or *sufficient*.

Squareness is a sufficient condition for rectangularity.

Being at least 35 years old is a necessary condition for being president of the United States.

Example. Rewrite the following as a universal conditional statement:

A product of two numbers is 0 only if one of the numbers is 0.

6.1: Set Theory: Definitions and the Element Method of Proof

Element Argument: The Basic Method for Proving that One set is a Subset of Another
Let sets X and Y be given. To prove that $X \subseteq Y$,

1. **suppose** that x is a particular but arbitrarily chosen element of X,
2. **show** that x is an element of Y

Example. Define sets A and B as follows:

$$A = \{m \in \mathbb{Z} \,|\, m = 6r + 12 \text{ for some } r \in \mathbb{Z}\}$$
$$B = \{n \in \mathbb{Z} \,|\, n = 3s \text{ for some } s \in \mathbb{Z}\}$$

Prove that $A \subseteq B$

Disprove that $B \subseteq A$

Definition.
Given sets A and B, A **equals** B, written $\mathbf{A = B}$, if, and only if, every element of A is in B and every element of B is in A:

$$A = B \Leftrightarrow A \subseteq B \text{ and } B \subseteq A.$$

Example. Define sets A and B as follows:

$$A = \{m \in \mathbb{Z} \,|\, m = 6r + 12 \text{ for some } r \in \mathbb{Z}\}$$
$$B = \{n \in \mathbb{Z} \,|\, n = 3s \text{ for some } s \in \mathbb{Z}\}$$

Is $A = B$?

Definition.
Given an integer n and a positive integer d, when n is divided by d, then

$$\begin{aligned} n \operatorname{div} d &= \text{ the integer quotient} \\ n \bmod d &= \text{ the nonnegative integer remainder} \end{aligned}$$

If n and d are integers and $d > 0$, then

$$n \operatorname{div} d = q \quad \text{and} \quad n \bmod d = r \quad \Leftrightarrow \quad n = dq + r$$

Example. Compute the following:

$32 \operatorname{div} 9, \;\; 32 \bmod 9$

$365 \operatorname{div} 7, \;\; 365 \bmod 7$

Example. If it is currently 11:00, what time will it be in

51 hours?

121 hours?

11 hours?

-1 hours?

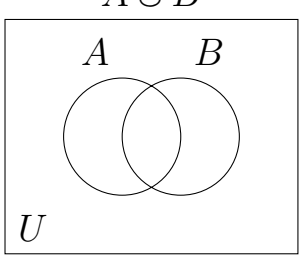
Example. Let $A = \left\{4, \sqrt{16}, 19 \bmod 15\right\}$ and $B = \{12 \bmod 8\}$. Is $A \subseteq B$? Is $B \subseteq A$?

Definition.
Let A and B be subsets of a universal set U.

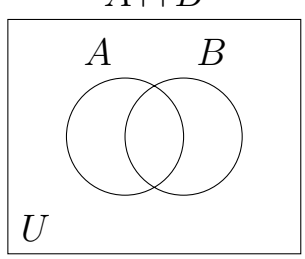
1. The **union** of A and B is the set of all elements that are in at least one of A or B.
$$A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$$
2. The **intersection** of A and B is the set of all elements that are common to both A and B.
$$A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$$
3. The **difference** of A and B is the set of all elements that are in B and not A.
$$B - A = \{x \in U \mid x \in B \text{ and } x \notin A\}$$
4. The **complement** of A is the set of all elements in U that are not in A.
$$A^c = \{x \in U \mid x \notin A\}$$

Example. Represent the following sets using the Venn diagrams below:

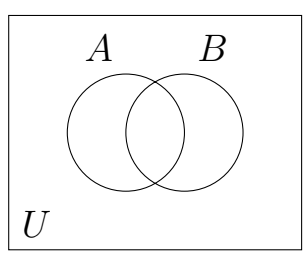
$A \cup B$



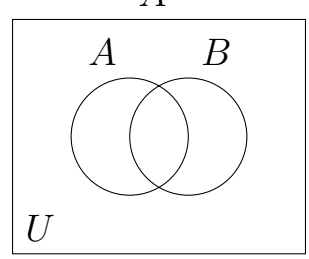
$A \cap B$



$B - A$



A^c



Example. Let the universal set be the set $U = \{a, b, c, d, e, f, g\}$, and let $A = \{a, c, e, g\}$ and $B = \{d, e, f, g\}$. Find

$A \cup B$

$A \cap B$

$B - A$

A^c

Definition.
Given real numbers a and b with $a \leq b$:

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\} \qquad (a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$$
$$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\} \qquad [a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$$

Example. Let the universal set be $\mathbb{R}$, and let $A = (-1, 0]$ and $B = [0, 1)$. Find

$A \cup B$

$A \cap B$

$B - A$

A^c

Definition.
Given sets $A_0, A_1, A_2, \ldots$ that are subsets of a universal set U and given a nonnegative integer n,

$$\bigcup_{i=0}^{n} A_i = \{x \in U \mid x \in A_i, \text{ for at least one } i = 0, 1, 2, \ldots, n\}$$

$$\bigcap_{i=0}^{n} A_i = \{x \in U \mid x \in A_i, \text{ for every } i = 0, 1, 2, \ldots, n\}$$

Example. For each positive integer i, let $A_i = \left\{x \in \mathbb{R} \,\middle|\, -\frac{1}{i} < x < \frac{1}{i}\right\} = \left(-\frac{1}{i}, \frac{1}{i}\right)$. Find

$A_1 \cup A_2 \cup A_3$

$A_1 \cap A_2 \cap A_3$

$\bigcup_{i=1}^{\infty} A_i$

$\bigcap_{i=1}^{\infty} A_i$

Definition.
The **empty set** (or **null set**), denoted $\varnothing$, is the set with no elements.

$$\{1, 3\} \cap \{2, 4\} = \varnothing$$

Two sets are called **disjoint** if, and only if, they have no elements in common:

$$A \cap B = \varnothing.$$

Sets $A_1, A_2, A_3, \ldots$ are **mutually disjoint** (or **pairwise disjoint**) if, and only if, no two sets A_i and A_j with distinct subscripts have any elements in common:

$$A_i \cap A_j = \varnothing \text{ whenever } i \neq j.$$

Example.
Let $A_1 = \{3, 5\}$, $A_2 = \{1, 4, 6\}$, and $A_3 = \{2\}$. Are A_1, A_2, and A_3 mutually disjoint?

Let $B_1 = \{2, 4, 6\}$, $B_2 = \{3, 7\}$, and $B_3 = \{4, 5\}$. Are B_1, B_2, B_3 mutually disjoint?

Definition.
A finite or infinite collection of nonempty sets $\{A_1, A_2, A_3, \ldots\}$ is a **partition** of a set A if, and only if,

1. A is the union of all the A_i;
2. the sets $A_1, A_2, A_3, \ldots$ are mutually disjoint.

Example.
Let $A = \{1, 2, 3, 4, 5, 6\}$, $A_1 = \{1, 2\}$, $A_2 = \{3, 4\}$, and $A_3 = \{5, 6\}$. Is $\{A_1, A_2, A_3\}$ a partition of A?

Let $\mathbb{Z}$ be the set of all integers and let

$$T_i = \{n \in \mathbb{Z} \mid n = 3k + i, \text{ for some integer } k\}.$$

Is $\{T_0, T_1, T_2\}$ a partition of $\mathbb{Z}$?

Definition.
Given a set A, the **power set** of A, denoted $\mathscr{P}(A)$, is the set of all subsets of A.

Example. Find $\mathscr{P}(\{x, y\})$.

6.4: Boolean Algebras, Russell's Paradox, and the Halting Problem

Definition.
A **Boolean algebra** is a set B together with two operations, generally denoted $+$ and $\cdot$, such that for all a and b in B both $a+b$ and $a \cdot b$ are in B and the following axioms are assumed to hold:

1. *Commutative Laws:* For all a and b in B,
$$a+b=b+a \text{ and } a \cdot b=b \cdot a$$
2. *Associative Laws:* For all a and b in B,
$$(a+b)+c=a+(b+c) \text{ and } (a \cdot b) \cdot c=a \cdot(b \cdot c)$$
3. *Distributive Laws:* For all a and b in B,
$$a+(b \cdot c)=(a+b) \cdot(a+c) \text{ and } a \cdot(b+c)=(a \cdot b)+(a \cdot c)$$
4. *Identity Laws:* There exist distinct elements 0 and 1 in B such that for each a in B,
$$a+0=a \text{ and } a \cdot 1=a$$
5. *Complement Laws:* For each a in B, there exists an element in B, denoted $\overline{a}$ and called the **complement** or **negation** of a, such that
$$a+\overline{a}=1 \text{ and } a \cdot \overline{a}=0$$

Properties of a Boolean Algebra
Let B be any Boolean algebra.

1. *Uniqueness of the Complement Laws:* For all a and x in B, if $a + x = 1$ and $a \cdot x = 0$, then $x = \overline{a}$.

2. *Uniqueness of* 0 *and* 1*:* If there exists x in B such that $a + x = a$ for every a in B, then $x = 0$, and if there exists y in B such that $a \cdot y = a$ for every a in B, then $y = 1$.

3. *Double Complement Law:* For every $a \in B$, $\overline{(\overline{a})} = a$.

4. *Idempotent Laws:* For every $a \in B$,
$$a + a = a \text{ and } a \cdot a = a.$$

5. *Universal Bound Laws:* For every $a \in B$,
$$a + 1 = 1 \text{ and } a \cdot 0 = 0.$$

6. *De Morgan's Laws:* For all a and $b \in B$,
$$\overline{a + b} = \overline{a} \cdot \overline{b} \text{ and } \overline{a \cdot b} = \overline{a} + \overline{b}.$$

7. *Absorption Laws:* For all a and $b \in B$,
$$(a + b) \cdot a = a \text{ and } (a \cdot b) + a = a.$$

8. *Complements of* 0 *and* 1*:*
$$\overline{0} = 1 \text{ and } \overline{1} = 0.$$

Example. Prove that for all elements a in a Boolean algebra B:

$\overline{(\overline{a})} = a.$

$a + a = a.$

Example. Prove that for all elements a in a Boolean algebra B:

$$a \cdot a = a.$$

$$(a + b) \cdot a = a.$$

Russell's Paradox
Define the following set S:

$$S = \{A \mid A \text{ is a set and } A \notin A\}.$$

Is S an element of itself?

The Barber Puzzle: In a certain town, there is a male barber who shaves all those men, and only those men, who do not shave themselves.
Does the barber shave himself?

Is the sentence "The barber shaves himself" a statement?

Example. Determine whether each sentence is a statement:

If $1 + 1 = 3$, then $1 = 0$.

This sentence is false and $1 + 1 = 3$.

The Halting Problem (Alan M. Turing)
There is no computer algorithm that will accept any algorithm X and data set D as input and then will output "halts" or "loops forever" to indicate whether or not X terminates in a finite number of steps when X is run with data set D.

Example boolean algebras:

Logical Equivalences	**Set Properties**
For all statement variables p, q, and r:	For all sets A, B, and C:
$p \vee q \equiv q \vee p$	$A \cup B = B \cup A$
$p \wedge q \equiv q \wedge p$	$A \cap B = B \cap A$
$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$	$A \cap (B \cap C) = (A \cap B) \cap C$
$p \vee (q \vee r) \equiv (p \vee q) \vee r$	$A \cup (B \cup C) = (A \cup B) \cup C$
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
$p \vee \mathbf{c} \equiv p$	$A \cup \varnothing = A$
$p \wedge \mathbf{t} \equiv p$	$A \cap U = A$
$p \vee \sim p \equiv \mathbf{t}$	$A \cup A^c = U$
$p \wedge \sim p \equiv \mathbf{c}$	$A \cap A^c = \varnothing$
$\sim (\sim p) \equiv p$	$(A^c)^c = A$
$p \wedge p \equiv p$	$A \cup A = A$
$p \vee p \equiv p$	$A \cap A = A$
$p \vee \mathbf{t} \equiv \mathbf{t}$	$A \cup U = U$
$p \wedge \mathbf{c} \equiv \mathbf{c}$	$A \cap \varnothing = \varnothing$
$\sim (p \vee q) \equiv \sim p \wedge \sim q$	$(A \cup B)^c = A^c \cap B^c$
$\sim (p \wedge q) \equiv \sim p \vee \sim q$	$(A \cap B)^c = A^c \cup B^c$
$p \vee (p \wedge q) \equiv p$	$A \cup (A \cap B) = A$
$p \wedge (p \vee q) \equiv p$	$A \cap (A \cup B) = A$
$\sim \mathbf{t} \equiv \mathbf{c}$	$U^c = \varnothing$
$\sim \mathbf{c} \equiv \mathbf{t}$	$\varnothing^c = U$

8.1: Relations on Sets

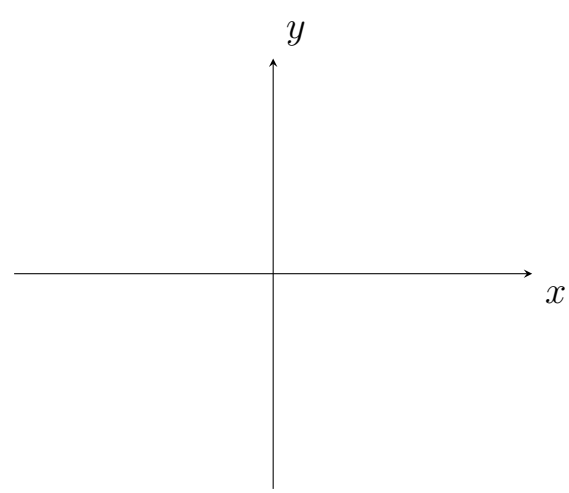
Definition.
A relation R from A to B is called a **binary relation** because it is a subset of a Cartesian product of two sets.

Example. Define a relation L from $\mathbb{R}$ to $\mathbb{R}$:

$$\forall x, y \in \mathbb{R},\ x\, L\, y \Leftrightarrow x < y.$$

Is $57\, L\, 53$? Is $(-17)\, L\, (-14)$? Is $143\, L\, 143$? Is $(-35)\, L\, 1$?

Draw the graph of L as a subset of the Cartesian plane $\mathbb{R} \times \mathbb{R}$.



Definition.
Two integers m and n are **congruent modulo** 2 if, and only if, $m \bmod 2 = n \bmod 2$.

Example. Define a relation E from $\mathbb{Z}$ to $\mathbb{Z}$:

$$\forall (m, n) \in \mathbb{Z} \times \mathbb{Z},\ m\, E\, n \Leftrightarrow m - n \text{ is even.}$$

Is $4\, E\, 0$? Is $2\, E\, 6$? Is $3\, E\, (-3)$? Is $5\, E\, 2$?

Prove that if n is any odd integer, then $n\, E\, 1$.

Example. Let $X = \{a, b, c\}$. Define a relation $\mathbf{S}$ from $\mathscr{P}(X)$ to $\mathscr{P}(X)$ as follows:

$$\forall A, B \in \mathscr{P}(X),\ A\, \mathbf{S}\, B \Leftrightarrow A \text{ has at least as many elements as } B.$$

Is $\{a, b\}\, \mathbf{S}\, \{b, c\}$?

Is $\{a\}\, \mathbf{S}\, \varnothing$?

Is $\{b, c\}\, \mathbf{S}\, \{a, b, c\}$?

Is $\{c\}\, \mathbf{S}\, \{a\}$?

Definition.
Let R be a relation from A to B. Define the inverse relation R^{-1} from B to A as follows:

$$R^{-1} = \{(y, x) \in B \times A \mid (x, y) \in R\}.$$

Example. Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$, and let R be the "divides" relation from A to B:

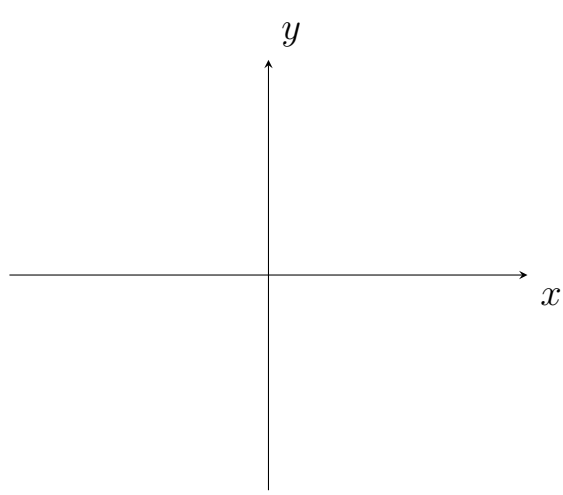
$$\forall (x, y) \in A \times B,\ x\, R\, y \Leftrightarrow x \mid y$$

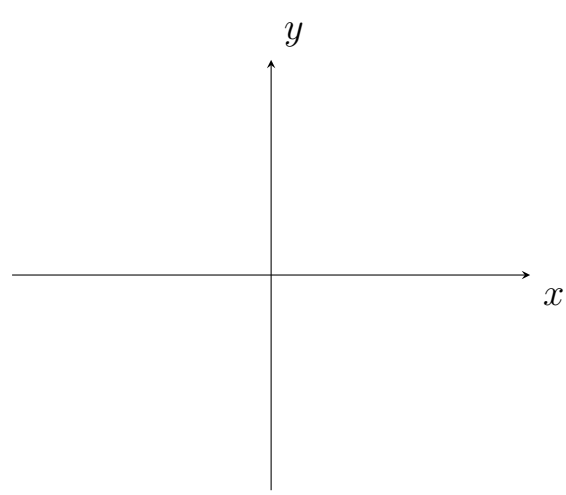
Explicitly state which ordered pairs are in R and R^{-1}. Draw arrow diagrams for both.

Example. Define a relation R from $\mathbb{R}$ to $\mathbb{R}$ as follows:

$$\forall (x, y) \in \mathbb{R} \times \mathbb{R},\ x\, R\, y \Leftrightarrow y = 2|x|.$$

Draw the graphs of R and R^{-1} in the Cartesian plane. Is R^{-1} a function?





Definition.
A **relation on a set** A is a relation from A to A.
A **graph** G consists of two finite sets:

- a nonempty set $V(G)$ of **vertices** and
- a set $E(G)$ of **edges**,

where each edge is associated with a set consisting of either one or two vertices called its endpoints.
A **directed graph** is a graph whose edges are directional.

Example. Let $A = \{3, 4, 5, 6, 7, 8\}$ and define a relation R on A as follows

$$\forall x, y \in A,\ x\, R\, y \Leftrightarrow 2 \mid (x - y).$$

Draw the directed graph of R.

Definition.
Given sets $A_1, A_2, \ldots, A_n$, an **n-ary relation** R on $A_1 \times A_2 \times \cdots \times A_n$ is a subset of $A_1 \times A_2 \times \cdots \times A_n$. The special case of 2-ary, 3-ary, and 4-ary relations are called **binary**, **ternary**, and **quaternary relations**, respectively.

8.2: Reflexivity, Symmetry, and Transitivity

Definition.
Let R be a relation on a set A.

1. R is **reflexive** if, and only if, for every $x \in A$, $x\,R\,x$.

$$\forall x \in A, (x, x) \in R$$

2. R is **symmetric** if, and only if, for every $x, y \in A$, if $x\,R\,y$ then $y\,R\,x$.

$$\forall x, y \in A, \text{ if } (x, y) \in R \text{ then } (y, x) \in R$$

3. R is **transitive** if, and only if, for every $x, y, z \in A$, if $x\,R\,y$ and $y\,R\,z$, then $x\,R\,z$.

$$\forall x, y, z \in A, \text{ if } (x, y) \in R \text{ and } (y, z) \in R \text{ then } (x, z) \in R$$

Note: A relation R is

not reflexive $\Leftrightarrow \exists x \in A$ such that $x \not R\, x$
or $(x, x) \notin R$.

not symmetric $\Leftrightarrow \exists x, y \in A$ such that $x\,R\,y$ but $y \not R\, x$
or $(x, y) \in R$ but $(y, x) \notin R$.

not transitive $\Leftrightarrow \exists x, y, z \in A$ such that $x\,R\,y$ and $y\,R\,z$, but $x \not R\, z$
or $(x, y) \in R$ and $(y, z) \in R$, but $(x, z) \notin R$.

irreflexive $\Leftrightarrow \forall x \in A, x \not R\, x$

asymmetric $\Leftrightarrow \forall x, y \in A$, if $x\,R\,y$ then $y \not R\, x$

intransitive $\Leftrightarrow \forall x, y, z \in A$, if $x\,R\,y$ and $y\,R\,z$, then $x \not R\, z$

Example. Define a relation R on $\mathbb{R}$ as follows:

$$x\,R\,y \Leftrightarrow x = y.$$

Is R reflexive? Is R symmetric? Is R transitive?

Example. Define a relation R on $\mathbb{R}$ as follows:

$$x\,R\,y \Leftrightarrow x < y.$$

Is R reflexive? Is R symmetric? Is R transitive?

Example. Let $A = \{0, 1, 2, 3\}$ and define relations R, S, and T on A as follows:

$$R = \{(0,0), (0,1), (0,3), (1,0), (1,1), (2,2), (3,0), (3,3)\}$$
$$S = \{(0,0), (0,2), (0,3), (2,3)\}$$
$$T = \{(0,1), (2,3)\}$$

For each relation, draw the directed graph, then identify if it is reflexive, symmetric, and/or transitive.

R

$0 \bullet$ $\bullet\ 1$

$3 \bullet$ $\bullet\ 2$

S

$0 \bullet$ $\bullet\ 1$

$3 \bullet$ $\bullet\ 2$

T

$0 \bullet$ $\bullet\ 1$

$3 \bullet$ $\bullet\ 2$

Example. Define a relation T on $\mathbb{Z}$ as follows:

$$\forall m, n \in \mathbb{Z},\ m\,T\,n \Leftrightarrow 3 \mid (m-n).$$

This relation is called **congruence modulo** 3**.**

Is T reflexive? Is T symmetric? Is T transitive?

Example. Define a relation S on $\mathbb{R}$ as follows:

$$\forall x, y \in \mathbb{R},\ x\, S\, y \Leftrightarrow |x| + |y| = 1.$$

Is S reflexive? Is S symmetric? Is S transitive?

8.3: Equivalence Relations

Definition.
Given a partition of a set A, the **relation induced by the partition**, R, is defined on A as follows:

$$\forall x, y \in A,\ x\,R\,y \ \Leftrightarrow \ \begin{array}{l}\text{there is a subset } A_i \text{ of the partition} \\ \text{such that both } x \text{ and } y \text{ are in } A_i.\end{array}$$

Example. Let $A = \{0, 1, 2, 3, 4\}$ and consider the following partition of A:

$$\{0, 3, 4\}, \{1\}, \{2\}.$$

Find the relation R induced by this partition

Let A be a set with a partition and let R be the relation induced by the partition. Then R is reflexive, symmetric, and transitive.

Definition.
Let A be a set and R a relation on A, R is an **equivalence relation** if, and only if, R is reflexive, symmetric, and transitive.

Example. Let X be the set of all nonempty subsets of $\{1, 2, 3\}$. Then

$$X = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Define a relation $\mathbf{R}$ on X as follows:

$$\forall A, B \in X,\ A\,\mathbf{R}\,B \Leftrightarrow \text{ the least element of } A \text{ equals the least element of } B.$$

Prove that $\mathbf{R}$ is an equivalence relation on X.

Definition.
Suppose A is a set and R is an equivalence relation on A. For each element a in A, the **equivalence class of** a, denoted $[a]$ and called the **class of** a, is the set of all elements x in A such that x is related to a by R:

$$[a] = \{x \in A | x \, R \, a\}.$$

Example. Let $A = \{0, 1, 2, 3, 4\}$ and define a relation R on A as follows:

$$R = \{\{0, 0\}, \{0, 4\}, \{1, 1\}, \{1, 3\}, \{2, 2\}, \{3, 1\}, \{3, 3\}, \{4, 0\}, \{4, 4\}\}.$$

Find the *distinct* equivalence classes of R.

Example. Recall the equivalence relation $\mathbf{R}$ defined on the nonempty subsets of $\{1, 2, 3\}$, X:

$$X = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

List the equivalence classes of $\mathbf{R}$.

Example. Consider the equivalence relation R on a set A:

$$\forall x, y \in A \, x \, R \, y \Leftrightarrow x = y.$$

Describe the distinct equivalence classes of R.

Suppose A is a set, R is an equivalence relation on A and a and b are elements of A.

- If $a \, R \, b$, then $[a] = [b]$.
- Either $[a] \cap [b] = \varnothing$ or $[a] = [b]$.

Example. Let R be the relation of congruence modulo 3 on the set $\mathbb{Z}$:

$$m\,R\,n \Leftrightarrow 3 \mid (m-n).$$

Describe the distinct equivalence classes of R.

Definition.
Let $m, n \in \mathbb{Z}$ and let $d \in \mathbb{Z}^+$. We say m **is congruent to** n **modulo** d

$$m \equiv n \mod d$$

if, and only if,

$$d \mid (m-n).$$

8.4: Modular Arithmetic with Applications to Cryptography

Definition.
The **Caesar cipher** encrypts messages by shifting letters by a fixed amount (typically 3 places). The encrypted messages are decrypted by shifting in the opposite direction.

Example. Use a Ceasar cipher with a shift of 3 to

encrypt the message MATH RULES

decrypt the message ODQGHU

Definition.
Given integers a and n with $n > 1$, the **residue of a modulo n** is $a \mod n$, the non-negative remainder obtained when a is divided by n.

- The numbers $0, 1, \ldots, n-1$ are called a **complete set of residues modulo n.**
- **Reducing a number modulo** n means to set it equal to its residue modulo n.
- If a modulus $n > 1$ is fixed throughout a discussion and an integer a is given, the words "modulo n" are often dropped and we simply speak of **the residue of a.**

Theorem 8.4.2: Congruence Modulo n is an Equivalence Relation
If n is any integer with $n > 1$, congruence modulo n is an equivalence relation on the set of all integers. The distinct equivalence classes of the relation are the sets $[0], [1], [2], \ldots, [n-1]$, where for each $a = 0, 1, 2, \ldots, n-1$,

$$\begin{aligned}[a] &= \{m \in \mathbb{Z} \mid m \equiv a \mod n\} \\ &= \{m \in \mathbb{Z} \mid m = a + kn \text{ for some integer } k\}\end{aligned}$$

Theorem 8.4.3: Modular Arithmetic
Let a, b, c, d, and n be integers with $n > 1$, and suppose

$$a \equiv c \mod n \qquad \text{and} \qquad b \equiv d \mod n.$$

Then

1. $(a + b) \equiv (c + d) \mod n$
2. $(a - b) \equiv (c - d) \mod n$
3. $ab \equiv cd \mod n$
4. $a^m \equiv c^m \mod n$, for every positive integer m.

Corollary 8.4.4
Let a, b and n be integers with $n > 1$. Then

$$ab \equiv [(a \bmod n)(b \bmod n)](\bmod n)$$

or, equivalently,

$$ab \mod n = [(a \bmod n)(b \bmod n)](\bmod n)$$

In particular, if m is a positive integer, then

$$a^m \equiv [(a \bmod n)^m](\bmod n).$$

Example. Compute the following

$55 + 26 \mod 4$

$55 - 26 \mod 4$

$55 \cdot 26 \mod 4$

$55^2 \mod 4$

Recall the following property of exponents:

- $x^{2a} = (x^a)^2$
- $x^{a+b} = x^a x^b$

for all real numbers x, a, and b with $x \geq 0$. By Corollary 8.4.4,

$$x^7 \mod n = \left[(x^4 \bmod n)(x^2 \bmod n)(x^1 \bmod n)\right] \mod n.$$

Example. Compute the following:

$144^4 \mod 713$ $\qquad$ $12^{43} \mod 713$

Definition.
An integer d is said to be a **linear combination of integers** a and b if, and only if, there exists integers s and t such that $as + bt = d$.

Theorem 8.4.5: Writing a Greatest Common Divisor as a Linear Combination
For all integers s and b, not both zero, if $d = \gcd(a, b)$, then there exist integers s and t such that $as + bt = d$.

Example. Use the extended Euclidean algorithm to express $\gcd(660, 43)$ as a linear combination of 660 and 43.

Definition.
Given any integer a and any positive integer n, if there exists an integer s such that $as \equiv 1(\mod n)$, then s is called **an inverse for a modulo n.**
Integers a and b are **relatively prime** if, and only if, $\gcd(a, b) = 1$.

Corollary 8.4.6
If a and b are relatively prime integers, then there exist integers s and t such that $as + bt = 1$.

Corollary 8.4.7
For all integers a and n, if $\gcd(a, n) = 1$, then there exists an integer s such that $as \equiv 1(\mod n)$, and so s is an inverse for a modulo n.

Example.

Find an inverse for 43 modulo 660. (Solve $43s \equiv 1(\mod 660)$.)

Find a positive inverse for 3 modulo 40. (Solve $3s \equiv 1(\mod 40)$.)

Definition. (RSA Cryptography)
RSA Encryption, named after Rivest, Shamir, and Adleman, uses modular arithmetic to encrypt and decrypt a plaintext message, M, into a ciphertext, C.
Given prime integers p and q, and a positive integer e that is relatively prime to $(p-1)(q-1)$,

- messages are encrypted using the formula:

$$C = M^e \mod pq.$$

  The number pair (pq, e) is a **public key**, which is shared.

- messages are decrypted using the formula:

$$M = C^d \mod pq$$

  The number pair (pq, d) is a **private key**. The integer d is a positive inverse to e modulo $(p-1)(q-1)$.

The security of RSA encryption relies on the difficulty of factoring very large numbers.

Example. Let $p = 5$ and $q = 11$. Since $(p-1)(q-1) = 40$, we may choose $e = 3$, which is relatively prime to 40.

Encrypt the plaintext "HI"

Decrypt the ciphertext "17 14"

Algorithm 1: Algorithm 8.4.1: Extended Euclidean Algorithm

Input : A, B [integers with $A > B > 0$]

$a := A,\ b := B,\ s := 1,\ t := 0,\ u := 0,\ v := 1$

while $b \neq 0$ **do**

$r := a \bmod b, \qquad q := a \operatorname{div} b$

$a := b, \qquad b := r$

$u_{\text{new}} := s - uq, \qquad v_{\text{new}} := t - vq$

$s := u, \qquad t := v$

$u := u_{\text{new}}, \qquad v := v_{\text{new}}$

end

gcd $:= a$

Output: gcd [a positive integer], s, t [integers]

8.5: Partial Order Relations

Definition.
Let R be a relation on a set A. R is **antisymmetric** if, and only if, for every a and b in A, if $a\,R\,b$ and $b\,R\,a$, then $a = b$.

Alternatively, if $a \neq b$, then either $a \not R\, b$ or $b \not R\, a$.

Example. Let R_1 and R_2 be relations on $\{0, 1, 2\}$ defined as follows:

$$R_1 = \{(0,2), (1,2), (2,0)\}, \qquad R_2 = \{(0,0), (0,1), (0,2), (1,1), (1,2)\}$$

Draw the directed graphs for R_1 and R_2 and indicate which relations are antisymmetric.

Definition.
Let R be a relation defined on a set A. R is a **partial order relation** if, and only if, R is reflexive, antisymmetric, and transitive.

Example. For non-empty sets U, and V, define the relation S such that

$$U\,S\,V \Leftrightarrow U \subseteq V$$

Prove that S is a partial order relation.

Example. For integers a, and b, define the relation D such that

$$a\,D\,b \Leftrightarrow a \mid b$$

Prove that D is a partial order relation.

Definition.
The symbol $\preceq$ is often used to denote a general partial order relation.

Lexicographic Order
Let A be a set with a partial order relation R, and let S be a set of strings over A. Define a relation $\preceq$ on S as follows:

> Let s and t be any strings in S of length m and n, respectively, where m and n are positive integers, and let s_m and t_m be the characters in the mth position for s and t, respectively.

1. If $m \leq n$ and the first m characters of s and t are the same, then $s \preceq t$.
2. If the first $k-1$ characters in s and t are the same, $s_k \, R \, t_k$, and $s_k \neq t_k$, then $s \preceq t$.
3. If λ is the null string, then $\lambda \preceq s$.

If no strings are related by $\preceq$ other than by these three conditions, then $\preceq$ is a partial order relation on S.
This partial order relation is called the **lexicographic order for** S that corresponds to the partial order R on A.

Example. Let $A = \{x, y\}$ and let R be the following partial order relation on A:

$$R = \{(x, x), (x, y), (y, y)\}$$

Is $x \preceq x$?

Is $x \preceq xx$?

Is $yx \preceq yxy$?

Is $xxxyyy \preceq xy$?

Is $x \preceq y$?

Is $\lambda \preceq xy$?

Is $xyy \preceq xyx$?

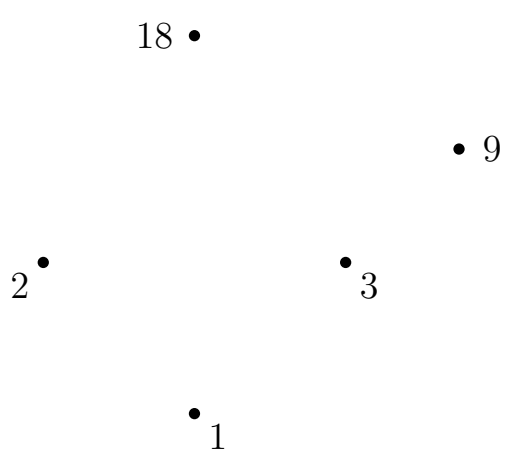
Definition.
A **Hasse diagram** is a graphical representation for a partial relation. To obtain a Hasse diagram, start with a directed graph of the relation with vertices placed so that all arrows point upwards. Then eliminate

- the loops at all vertices,
- all arrows implied by the transitive property, and
- the direction indicators on the arrows.

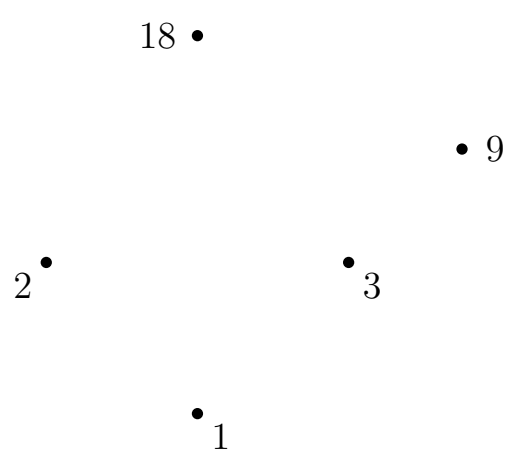
Example. Let $A = \{1, 2, 3, 9, 18\}$. Consider the partial order relation "divides":

$$a \, D \, b \Leftrightarrow a \mid b$$

Draw the directed graph of D:



Generate the Hasse diagram of D:



Example. Consider the "subset" relation, $\subseteq$, on the set $\mathscr{P}(\{a, b, c\})$:

$$U \, R \, V \Leftrightarrow U \subseteq V$$

Construct the Hasse diagram for this relation.

Definition.
Suppose $\preceq$ is a partial order relation on a set A. Elements a and b of A are said to be **comparable** if, and only if, either $a \preceq b$ or $b \preceq a$. Otherwise, a and b are called **noncomparable**.

When all the elements of a partial order relation are comparable, the relation is called a **total order**.

If R is a partial order relation on a set A, and for any two elements a and b in A either $a\,R\,b$ or $b\,R\,a$, then R is a **total order relation** on A.

A set A is called a **partially ordered set** (or **poset**) with respect to a relation $\preceq$ if, and only if, $\preceq$ is a partial order relation on A.

A set A is called a **totally ordered set** with respect to a relation $\preceq$ if, and only if A is partially ordered with respect to $\preceq$ and $\preceq$ is a total order.

A subset B of A is called a **chain** if, and only if, the elements in each pair of elements in B are comparable. The **length of a chain** is one less than the number of elements in the chain.

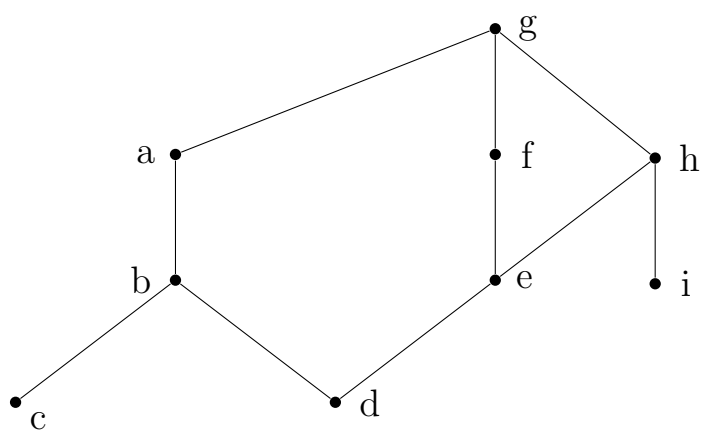
Note: The Hasse diagram for a total order relation can be drawn as a single vertical chain.

Example. The set $\mathscr{P}(\{a, b, c\})$ is partially ordered with respect to the subset relation. Find a chain of length 3 in $\mathscr{P}(\{a, b, c\})$.

Definition.
Let a set A be partially ordered with respect to a relation $\preceq$. An element in A is called a

1. **maximal element of** $A \Leftrightarrow \forall b \in A$, either $b \preceq a$ or b and a are not comparable.
2. **greatest element of** $A \Leftrightarrow \forall b \in A$, $b \preceq a$.
3. **minimal element of** $A \Leftrightarrow \forall b \in A$, either $a \preceq b$ or b and a are not comparable.
4. **least element of** $A \Leftrightarrow \forall b \in A$, $a \preceq b$.

Example. Let $A = \{a, b, c, d, e, f, g, h, i\}$ have the partial ordering $\preceq$ defined by the following Hasse diagram. Find all maximal, minimal, greatest, and least elements of A.



1.4: The Language of Graphs

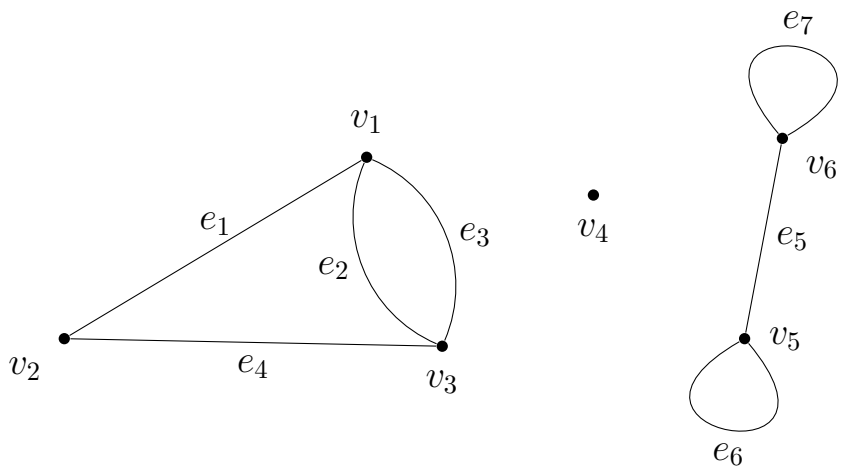
Definition.
A **graph** G consists of two finite sets:

- a nonempty set $V(G)$ of **vertices** and
- a set $E(G)$ of **edges**,

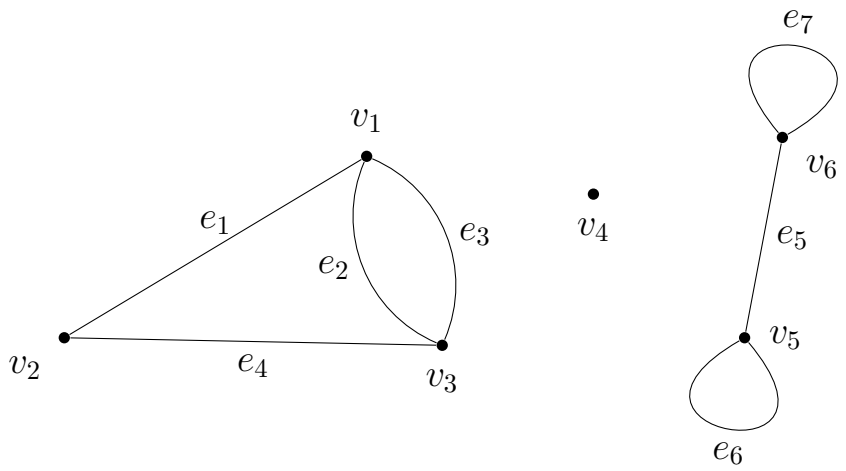
where each edge is associated with a set consisting of either one or two vertices called its **endpoints**.
The correspondence from edges to endpoints is called the **edge-endpoint function**.

- An edge with just one endpoint is called a **loop**.
- Two or more distinct edges with the same set of endpoints are said to be **parallel**.
- An edge is said to **connect** its endpoints.
- Two vertices that are connected by an edge are called **adjacent**.
- A vertex that is an endpoint of a loop is said to be **adjacent to itself**.
- An edge is said to be **incident on** each of its endpoints.
- Two edges incident on the same endpoint are called **adjacent**.
- A vertex on which no edges are incident is called **isolated**.

Example. Consider the following graph:



Write the vertex set, the edge set, and give a table showing the edge-endpoint function.



Find all

edges that are incident on v_1

vertices that are adjacent to v_1

edges that are adjacent to e_1

loops

parallel edges

vertices that are adjacent to themselves

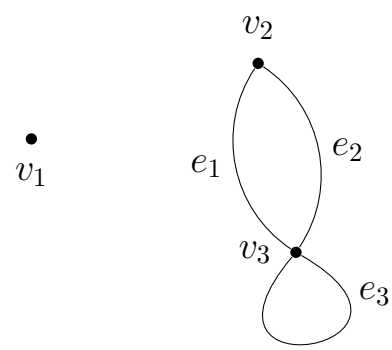
isolated vertices

Definition.
Let G be a graph and v a vertex of G. A **directed graph**, or **digraph**, consists of two finite sets:

- a nonempty set $V(G)$ of vertices and
- a set $D(G)$ of directed edges,

where each is associated with an ordered pair of vertices called its **endpoints**.
If edge e is associated with the pair (v, w) of vertices, then e is said to be the (**directed**) **edge** from v to w.
The degree of v, denoted $\deg(v)$, equals the number of edges that are incident on v, with an edge that is a loop counted twice.

Example. Find the degree of each vertex of the graph G shown below.



4.9: Application: The Handshake Theorem

Definition.
The total degree of a graph is the sum of the degrees of all the vertices of the graph.

Theorem 4.9.1: The Handshake Theorem
If G is any graph, then the sum of the degrees of all the vertices of G equals twice the number of edges of G. Specifically, if the vertices of G are $v_1, v_2, \ldots, v_n$, where n is a positive integer, then

$$\begin{aligned} \text{the total degree of } G &= \deg(v_1) + \deg(v_2) + \cdots + \deg(v_n) \\ &= 2 \cdot (\text{the number of edges of } G) \end{aligned}$$

Corollary 4.9.2
The total degree of a graph is even.

Example. Draw a graph with the specified properties, or show that no such graph exists

A graph with four vertices of degrees 1, 1, 2, and 3

A graph with four vertices of degrees 1, 1, 3, and 3

Proposition 4.9.3
In any graph there is an even number of vertices of odd degree.

Example. Is there a graph with 10 vertices of degrees $1, 1, 2, 2, 2, 3, 4, 4, 4$, and 6?

Example. At a meeting attended by a group of people, four people shook hands with one other person, six people shook hands with two other people, two people shook hands with three other people, and the rest shook hands with four other people. No two people shook hands with each other more than once, and a total of 23 handshakes occurred. How many people attended the meeting?

10.2: Matrix Representations of Graphs

Definition.
An $m \times n$ **matrix** A **over a set** S is a rectangular array of elements of S arranged into m rows and n columns:

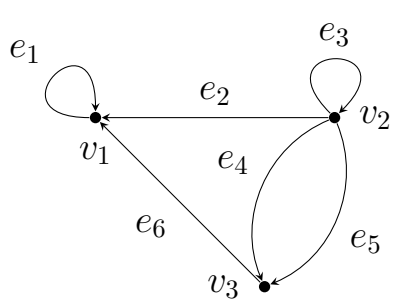
$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{jn} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}$$

Note: Two matrices A and B are only equal if every component of A and B are equal:

$$A = B \Leftrightarrow a_{ij} = b_{ij} \ \ \forall i = 1, 2, \dots, m, \text{ and } \forall j = 1, 2, \dots, n.$$

Definition.
Let G be a directed graph with ordered vertices $v_1, v_2, \dots, v_n$. The **adjacency matrix of** G is the $n \times n$ matrix $A = (a_{ij})$ over the set of nonnegative integers such that

$$a_{ij} = \text{ the number of arrows from } v_i \text{ to } v_j, \ \forall i, j = 1, 2, \dots, n.$$



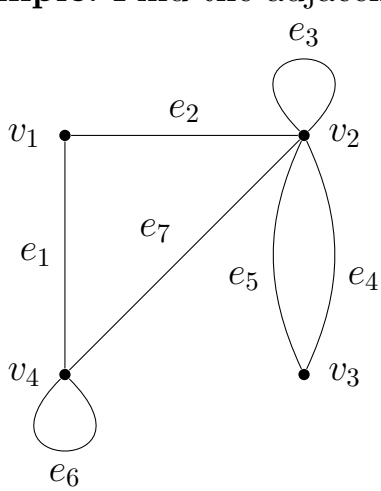
$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 2 \\ 1 & 0 & 0 \end{bmatrix} \end{array}$$

Example. Let

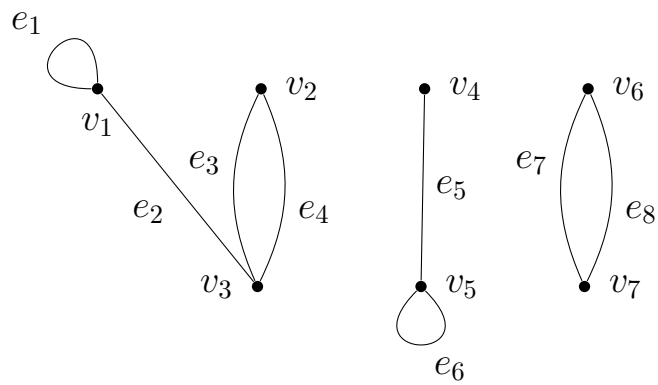
$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 2 & 1 & 0 & 0 \end{bmatrix}$$

Draw a directed graph that has A as its adjacency matrix.

Example. Find the adjacency matrix for the following graph:



Example. Find the adjacency matrix of the following disconnected graph:



Theorem 10.2.1
Let G be a graph with connected components $G_1, G_2, \ldots, G_k$. If there are n_i vertices in each connected component G_i and these vertices are numbered consecutively, then the adjacency matrix of G has the form

$$\begin{bmatrix} A_1 & O & O & \ldots & O \\ O & A_2 & O & \ldots & O \\ O & O & A_3 & \ldots & O \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ O & O & O & \ldots & A_k \end{bmatrix}$$

where each A_i is the $n_i \times n_i$ adjacency matrix of G_i, $\forall i = 1, 2, \ldots, k$, and the O's represent matrices whose entries are all 0.

Definition.
Suppose that all elements in matrices A and B are real numbers. If the number of elements n in the ith row of A equals the number of elements in the jth column of B, then the **scalar product** or **dot product** of the ith row of A and the jth column of B is the real number obtained as follows:

$$\begin{bmatrix} a_{i1} & a_{i2} & \dots & a_{in} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{bmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

Example. Multiple the following matrices:

$$\begin{bmatrix} -2 & -3 & 4 & 5 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \\ -3 \end{bmatrix}$$

Definition.
Let $A = (a_{ij})$ be an $m \times k$ matrix, and $B = (b_{ij})$ a $k \times n$ matrix with real entries. The matrix product of A times B, denoted AB is the matrix (c_{ij}) defined as:

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj} = \sum_{r=1}^{k} a_{ir}b_{rj}$$

for each $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$.

Example. Multiply the following matrices:

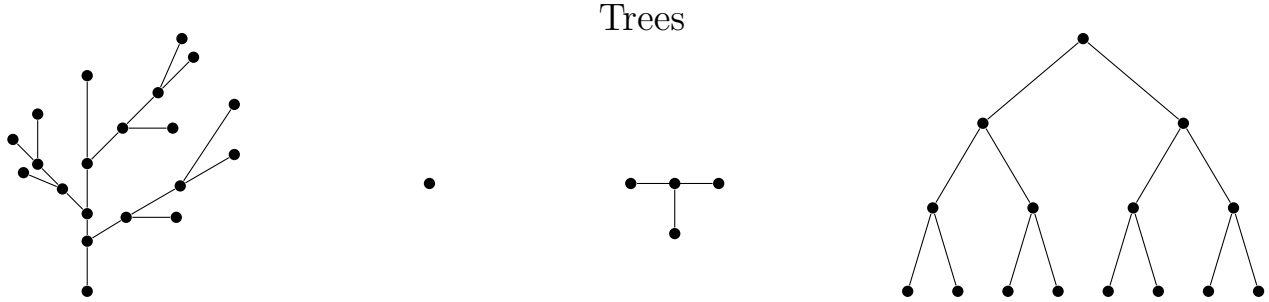
$$\begin{bmatrix} 2 & 0 & -1 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} 5 & 5 \\ -2 & -3 \\ 3 & -4 \end{bmatrix}$$

10.4: Trees: Examples and Basic Properties

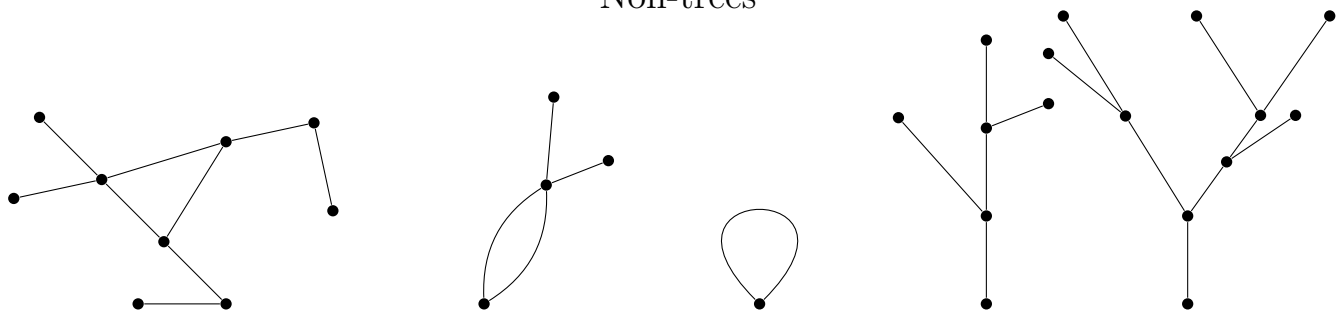
Definition.

- A **closed walk** is a finite alternating sequence of adjacent vertices and edges that starts and ends at the same vertex.
- A **circuit** is a closed walk that contains at least one edge and does not contain a repeated edge.
- A graph is said to be **circuit-free** if, and only if, it has no circuits.
- A graph is called a **tree** if, and only if, it is circuit-free and connected.
- A **trivial tree** is a graph that consists of a single vertex.
- A graph is called a **forest** if, and only if, it is circuit-free and not connected.

Trees



Non-trees



Lemma 10.4.1
Any tree that has more than one vertex has at least one vertex of degree 1.

Definition.
Let T be a tree. If T has at least two vertices, then

- a vertex of degree 1 in T is called a **leaf** (or a **terminal vertex**), and
- a vertex of degree greater than 1 in T is called an **internal vertex** (or a **branch vertex**).

The unique vertex in a trivial tree is also called a **leaf** or **terminal vertex**.

Theorem 10.4.2
For any positive integer n, any tree with n vertices has $n-1$ edges.

Theorem 10.4.4
For any positive integer n, if G is a connected graph with n vertices and $n-1$ edges, then G is a tree.

Corollary 10.4.5
If G is any graph with n vertices and m edges, where m and n are positive integers and $m \geq n$, then G has a circuit.

Example. A graph G has ten vertices and twelve edges. Is it a tree?

Example. For the following, either draw a graph matching the description given, or justify why no such graph exists:

Tree, nine vertices, nine edges

Graph, connected, nine vertices, nine edges

Graph, circuit-free, nine vertices, six edges

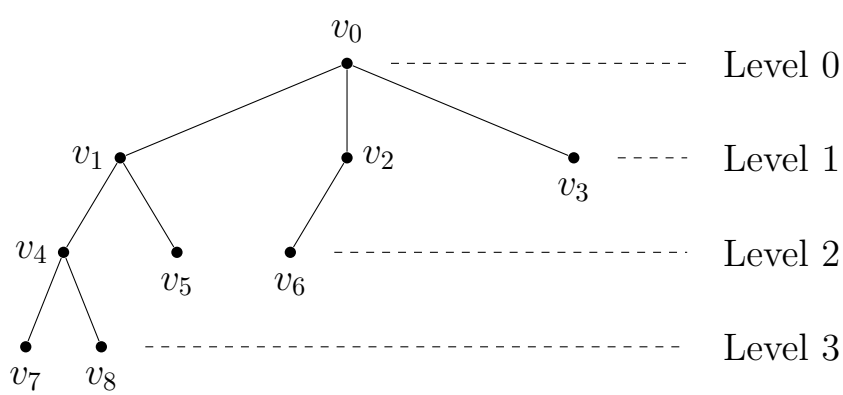
Tree, six vertices, total degree 14

Tree, five vertices, total degree 8

10.5: Rooted Trees

Definition.

- A **rooted tree** is a tree in which there is one vertex that is distinguished from the others and is called the **root**.
- The **level** of a vertex is the number of edges along the unique path between it and the root.
- The **height** of a rooted tree is the maximum level of any vertex of the tree.
- Given the root or any internal vertex v of a rooted tree, the **children** of v are all those vertices that are adjacent to v and are one level farther away from the root than v.
- If w is a child of v, then v is called the **parent** of w, and two distinct vertices that are both children of the same parent are called **siblings**.
- Given two distinct vertices v and w, if v lies on the unique path between w and the root, then v is an **ancestor** of w and w is a **descendant** of v.



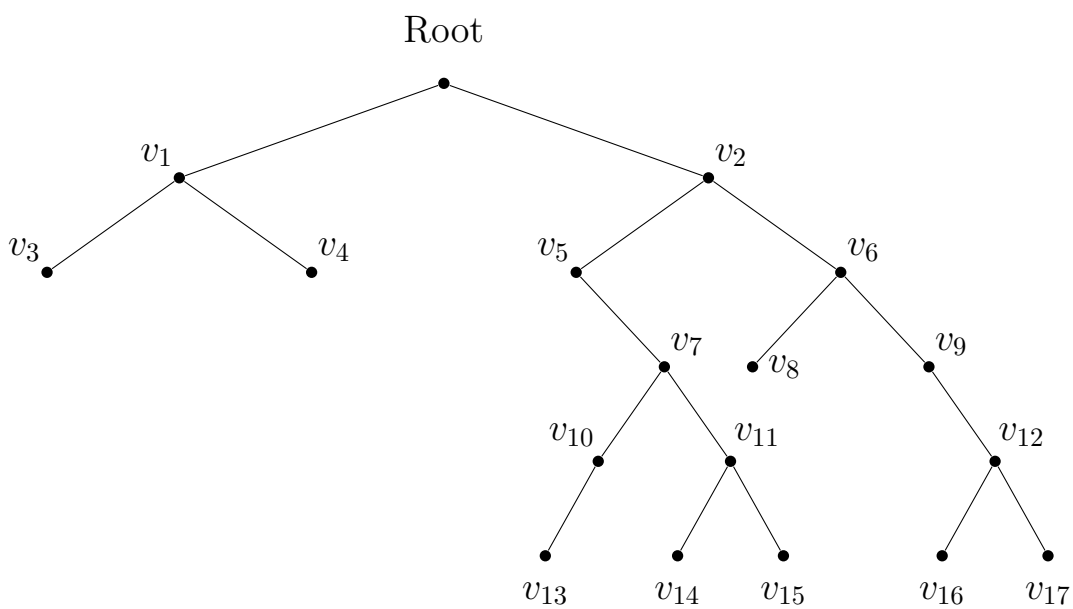
Definition.

- A **binary tree** is a rooted tree in which every parent has at most two children.
- Each child in a binary tree is designated either a **left child** or a **right child** (but not both), and every parent has at most one left child and one right child.
- A **full binary tree** is a binary tree in which each parent has exactly two children.

Given any parent v in a binary tree T, if v has a left child, then the **left subtree** of v is the binary tree

- whose root is the left child of v,
- whose vertices consist of the left child of v and all its descendants, and
- whose edges consist of all those edges of T that connect the vertices of the left subtree.

The **right subtree** of v is defined analogously.



Theorem 10.5.1
If k is a positive integer and T is a full binary tree with k internal vertices, then

- T has a total of $2k + 1$ vertices, and
- T has $k + 1$ leaves.

Theorem 10.5.2
For ever integer $h \geq 0$, if T is any binary tree with height h and t leaves, then

$$t \leq 2^h \Leftrightarrow \log_2(t) \leq h.$$

Corollary 10.5.3
A full binary tree of height h has 2^h leaves.

Example. Determine if the following binary trees can exist:

Binary tree, height 4, eight leaves

Full binary tree, nine vertices, five internal vertices

Full binary tree, four internal vertices

Full binary tree, height 4, eighteen leaves

Definition.

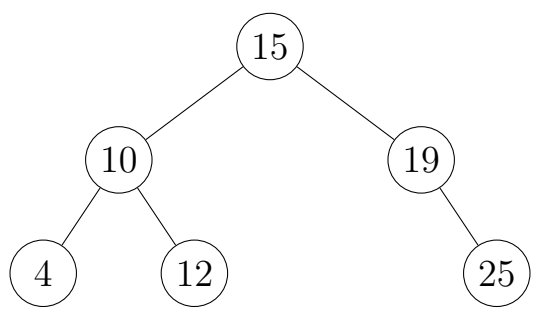
- A **binary search tree** is a binary tree in which data records can be stored, searched, and processed.
- Each element is assigned a **key**, which is used to order the elements.

In a binary tree, for every internal vertex v:

- all the keys in the left subtree of v are less than the key in v, and
- all the keys in the right subtree of v are greater than the key in v.

Example. The following is a binary search tree for the following keys:

$$15, 10, 19, 25, 12, 4$$



Example. Build the binary search tree for the keys $19, 10, 25, 4, 15, 12$

Algorithm 2: Algorithm 10.5.1: Building a Binary Search Tree

```
Input  : A totally ordered, nonempty set K of keys
Initialize T to have one vertex, the root, and no edges
Choose a key from K to insert into the root
while (there are still keys to be added) do
    Choose a key, newkey, from K to add.
    Let the root be called v, let key(v) be the key at the root
    success := 0
    while success == 0 do
        if newkey < key(v) then
            if v has a left child then
                v_L :=the left child
                v   := v_L
            else
                add a vertex v_L to T as the left child for v
                add an edge to T to join v to v_L
                insert newkey as the key for v_L
                success := 1
            end
        if newkey > key(v) then
            if v has a right child then
                v_R :=the right child
                v   := v_R
            else
                add a vertex v_R to T as the right child for v
                add an edge to T to join v to v_R
                insert newkey as the key for v_R
                success := 1
            end
    end
end
Output: A binary search tree T for the set K of keys
```

10.6: Spanning Trees and a Shortest Path Algorithm

Definition.
A **spanning tree** for a graph G is a subgraph of G that contains every vertex of G and is a tree.

Proposition 10.6.1

1. Every connected graph has a spanning tree.
2. Any two spanning trees for a graph have the same number of edges.

Definition.

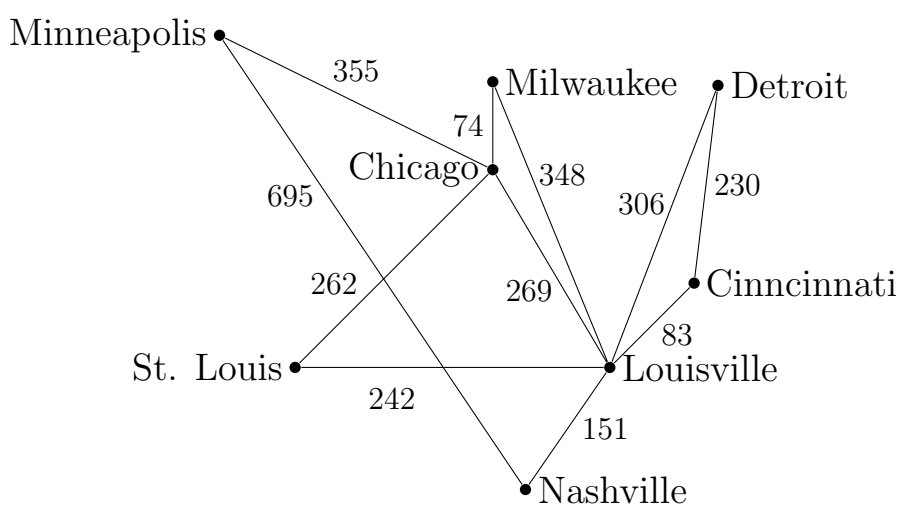
- A **weighted graph** is a graph for which each edge has an associated positive real number **weight**.
- The sum of the weights of all the edges is the **total weight** of the graph.
- A **minimum spanning tree** for a connected, weighted graph is a spanning tree that has the least possible total weight compared to all other spanning trees for the graph.

If G is a weighted graph and e is an edge of G, then $w(e)$ denotes the weight of e and $w(G)$ denotes the total weight of G.

Algorithm 3: Algorithm 10.6.1: Kruskal

Input : G [a connected, weighted graph with n vertices]
Initialize T to have all the vertices of G and no edges.
Let E be the set of all the edges of G, and let $m := 0$
while $m < n - 1$ **do**
  Find an edge e in E of least weight
  Delete e from E
  if *adding e to T does not produce a circuit* **then**
    add e to the edge set of T
    $m := m + 1$
end
Output: T [a minimum spanning tree for G]

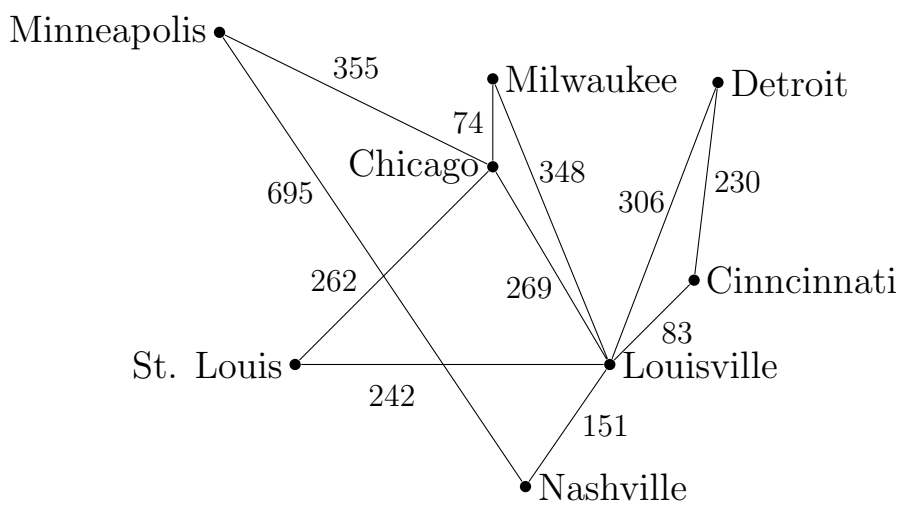
Example. Use Kruskal's algorithm to find a minimum spanning tree of the following graph:



Algorithm 4: Algorithm 10.6.2: Prim's Algorithm

Input : G [a connected, weighted graph with n vertices]
Pick a vertex v of G and let T be the graph with one vertex, v, and no edges
Let V be the set of all the vertices of G except v
for $i := 1$ ***to*** $n - 1$ **do**
  Find an edge e of G such that
    1. e connects T to one of the vertices in V, and
    2. e has the least weight of all edges connecting T to a vertex in V
  Let w be the endpoint of e that is in V.
  Add e and w to the edge and vertex sets of T, and delete w from V
end
Output: T [a minimum spanning tree for G]

Example. Use Prim's algorithm to find a minimum spanning tree of the following graph:



7.1: Functions Defined on General Sets

Definition.
A **function** f **from a set** X **to a set** Y, denoted $f : X \to Y$, is a relation from X, the **domain** of f, to Y, the **co-domain** of f, that satisfies

1. every element in X is related to some element in Y, and
2. no element in X is related to more than one element in Y.

Then, given any element $x \in X$, there is a unique element in Y that is related to x by f:

$$\forall x \in X,\ f(x) \in Y$$

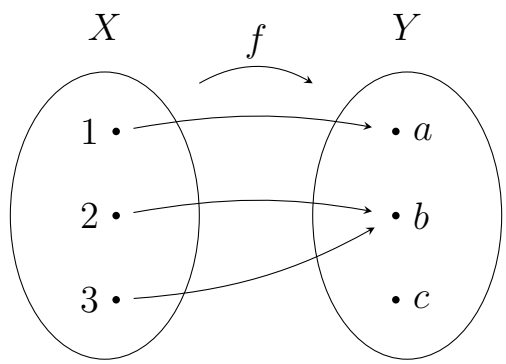
The set of all values of f taken together is called the **range of** f or the **image of** X **under** f:

$$\text{range of } f = \{y \in Y \mid y = f(x), \quad \text{for some } x \in X\}.$$

Given an element $y \in Y$, there may exist elements in X with y as their image. When x is an element such that $f(x) = y$, then x is called a **a preimage of** y or **an inverse image of** y:

$$\text{the inverse image of } y = \{x \in X \mid f(x) = y\}.$$

Example. Using the arrow diagram below, identify the following:



domain and co-domain

range

$f(1)$ $f(2)$ $f(3)$

Is 2 an inverse image of b? Is 2 an inverse image of c?

Find the inverse image of a, b, and c

Represent f as a set of ordered pairs

Example. Given a set X, define a function $I_X : X \to X$ (**identity function on** X):

$$I_X(x) = x, \ \forall x \in X$$

What are the domain, co-domain, and range of I_X?

Definition.

For $b > 0$ and $b \neq 1$, the **logarithmic function**

$$y = \log_b(x) \qquad \text{(logarithmic form)}$$

has domain $x > 0$, base b, and is defined by

$$b^y = x \qquad \text{(exponential form)}$$

Example. Find the following:

$\log_3(9)$ $\qquad$ $\log_2\left(\frac{1}{2}\right)$ $\qquad$ $\log_{10}(1)$

$\log_2\left(2^m\right)$ $\qquad$ $2^{\log_2(m)}, \ (m > 0)$

Definition.
If $f : X \to Y$ is a function and $A \subseteq X$ and $C \subseteq Y$, then

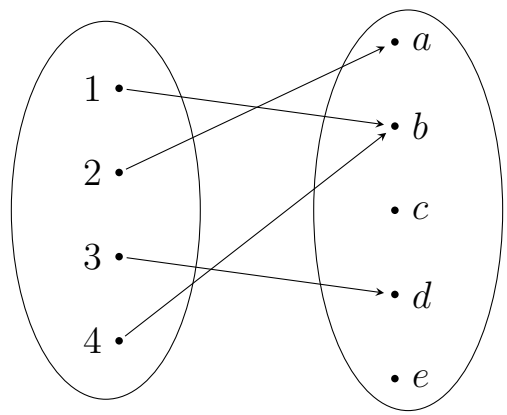
$$f(A) = \{y \in Y \mid y = f(x) \text{ for some } x \in A\}$$

and

$$f^{-1}(C) = \{x \in X \mid f(x) \in C\}.$$

$f(A)$ is called the **image of** A, and $f^{-1}(C)$ is called the **inverse image of** C.

Example. Let $X = \{1, 2, 3, 4\}$ and $Y = \{a, b, c, d, e\}$, and define $F : X \to Y$ by the following arrow diagram:



Let $A = \{1, 4\}$, $C = \{a, b\}$, and $D = \{c, e\}$. Find

$F(A)$ $\qquad\qquad$ $F(X)$

$F^{-1}(C)$ $\qquad\qquad$ $F^{-1}(D)$

7.2: One-to-One, Onto, and Inverse Functions

Definition.
Let F be a function from a set X to a set Y. F is **one-to-one** (or **injective**) if, and only if, for all elements x_1 and x_2 in X,

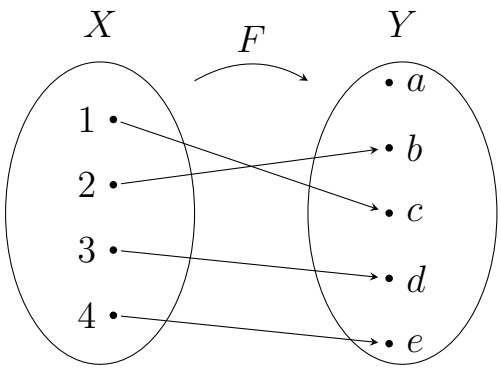
$$\text{if } F(x_1) = F(x_2), \text{ then } x_1 = x_2,$$

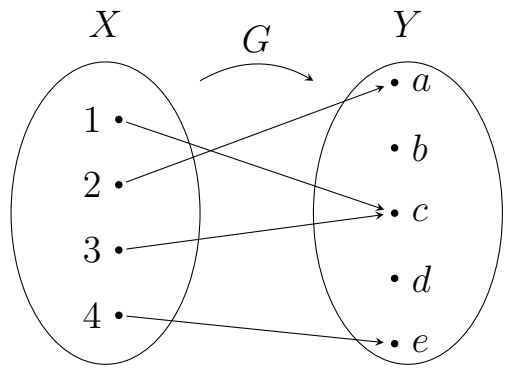
or, equivalently,

$$\text{if } x_1 \neq x_2, \text{ then } F(x_1) \neq F(x_2).$$

$$F : X \to Y \text{ is one-to-one } \Leftrightarrow \forall x_1, x_2 \in X, \text{ if } F(x_1) = F(x_2) \text{ then } x_1 = x_2.$$

Example. Do either of the arrow diagrams below define a one-to-one function?





Example. For the following functions, prove they are one-to-one, or show that they are not:

$$f(x) = 4x + 2$$

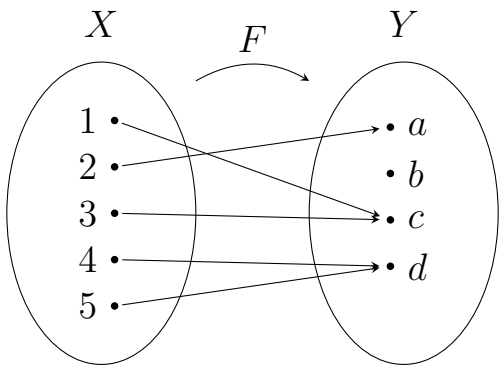
$$g(s) = \frac{s+2}{s-3}$$

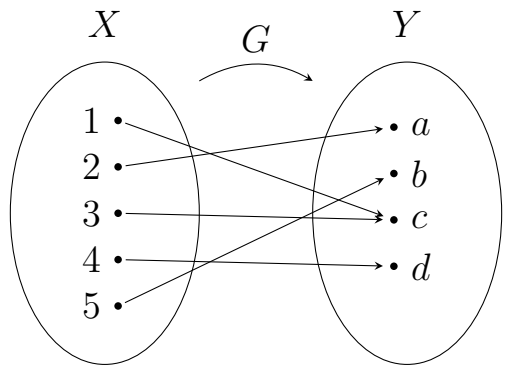
$$h(t) = \frac{e^t + e^{-t}}{2}$$

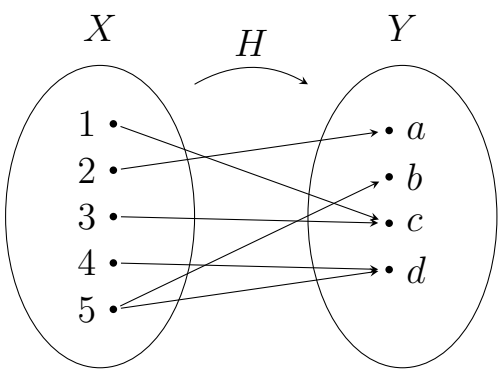
Definition.
Let F be a function from a set X to a set Y. F is **onto** (or **surjective**) if, and only if, given any element y in Y, it is possible to find an element x in X with the property that $y = F(x)$

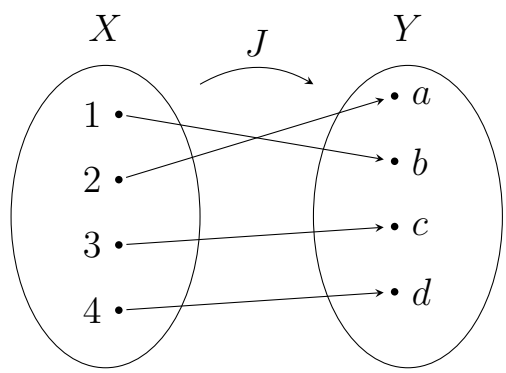
$$F : X \to Y \text{ is onto } \Leftrightarrow \forall y \in Y,\ \exists x \in X \text{ such that } F(x) = y.$$

Example. Do any of the arrow diagrams below define a function that is onto?









Example. For the following functions, prove they are onto, or show that they are not:

$f : \mathbb{R} \to \mathbb{R}$ where $f(x) = 4x - 1,\ \forall x \in \mathbb{R}$

$g : \mathbb{Z} \to \mathbb{Z}$ where $g(n) = 4n - 1,\ \forall n \in \mathbb{Z}$

Definition.
A **one-to-one correspondence** (or **bijection**) from a set X to a set Y is a function $F : X \to Y$ that is *both* one-to-one and onto.

If F is a one-to-one correspondence, then F^{-1} exists:

$$F(x) = y \Leftrightarrow F^{-1}(y) = x$$

Example. Find the inverse of the following functions:

$f(x) = 4x + 2$

$g(s) = \dfrac{s+2}{s-3}$

5.1: Sequences

Definition.
A **sequence** is a function whose domain is either all the integers between two given integers or all the integers greater than or equal to a given integer.

Example. Given

$$a_k = \frac{k}{k+1},\ k \in \mathbb{Z}^{\geq 1} \qquad\qquad b_i = \frac{i-1}{i},\ i \in \mathbb{Z}^{\geq 2}$$

Compute the first five terms of both sequences

Example. Find an explicit formula for a sequence with the following initial terms:

$$1,\ -\frac{1}{4},\ \frac{1}{9},\ -\frac{1}{16},\ \frac{1}{25},\ -\frac{1}{36},\ \ldots$$

Definition.
Let $m, n \in \mathbb{Z}$ with $m \leq n$. The **summation** of a_k from m to n is given by:

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \cdots + a_{n-1} + a_n$$

where k is the **index**, m is the **lower limit**, and n is the **upper limit**.

Example. Rewrite the following using summation notation:

$$1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \frac{1}{25} - \frac{1}{36}$$

Example. Compute $\sum_{k=1}^{5} k^2$

Example. Write $\sum_{k=1}^{n} 2^k + 2^{n+1}$ as a single summation

Example. Rewrite $\sum_{i=1}^{n+1} \frac{1}{i^2}$ by separating off the final term

Example. Evaluate the following telescoping sum:

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1} \right)$$

Definition.
Let $m, n \in \mathbb{Z}$ with $m \leq n$. The **product** of a_k from m to n is given by:

$$\prod_{k=m}^{n} a_k = a_m \cdot a_{m+1} \cdot a_{m+2} \cdot \cdots \cdot a_{n-1} \cdot a_n$$

where k is the **index**, m is the **lower limit**, and n is the **upper limit**.

Example. Compute the following:

$$\prod_{k=1}^{5} k \qquad\qquad \prod_{k=1}^{10} \frac{k}{k+1}$$

Example. Rewrite the following summation so that the index begins at $k = 1$:

$$\sum_{k=0}^{6} \frac{1}{k+1}$$

Definition.
For each positive integer n, the quantity n **factorial**, denoted $n!$, is defined to be the product of all of the integers from 1 to n:

$$n! = n \cdot (n-1) \ldots 3 \cdot 2 \cdot 1 = \prod_{k=1}^{n} k$$

Zero factorial, denoted $0!$, is defined to be 1:

$$0! = 1$$

Example. Simplify the following expressions:

$$\frac{8!}{7!} \qquad \frac{5!}{2! \cdot 3!}$$

$$\frac{(n+1)!}{n!} \qquad \frac{n!}{(n-3)!}$$

$$\frac{1}{2! \cdot 4!} + \frac{1}{3! \cdot 3!}$$

Definition.
Let $n, r \in \mathbb{Z}$ with $0 \leq r \leq n$. Then

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

is n **choose** r, which represents the number of subsets of size r that can be chosen from a set with n elements.

5.6: Defining Sequences Recursively

Definition.
A **recurrence relation** for a sequence $a_0, a_1, a_2, \ldots$ is a formula that relates each term a_k to a certain subset of its predecessors $a_{k-1}, a_{k-2}, \ldots, a_{k-i}$, where i is an integer with $k - i \geq 0$.
If i is a fixed integer, the **initial conditions** for such a recurrence relation specify the values $a_0, a_1, a_2, \ldots, a_{i-1}$.

Example. Define a sequence $c_0, c_1, c_2, \ldots$ recursively as follows: For every integer $k \geq 2$,

$$c_k = c_{k-1} + kc_{k-2} + 1$$
$$c_0 = 1 \text{ and } c_1 = 2$$

Find the first 5 elements of this sequence.

Example. The **Fibonacci Numbers** are a sequence commonly attributed to Leonardo of Pisa (a.k.a Fibonacci). He posed the following problem:

> A single pair of rabbits (male and female) is born at the beginning of a year. Assume the following conditions:
>
> 1. Rabbit pairs are not fertile during their first month of life but thereafter give birth to one new male/female pair at the end of every month.
> 2. No rabbits die.
>
> How many rabbits will there be at the end of the year?

Example. The **Tower of Hanoi** is a puzzle with 8 circular disks of wood with holes in the center. The disks are stacked from the smallest down to the largest. The objective of the game is to move the stack, one disk at a time, from one pole to another without placing a larger disk above a smaller disk:

Tower of Hanoi

What is the minimum number of moves required to solve this puzzle?

Example. Use mathematical induction to prove that for each integer $n \geq 0$,

$$F_{n+2}F_n - F_{n+1}^2 = (-1)^n$$

Example. Consider the following recursive sequence:

$$x_n = \sqrt{4 + 3x_{n-1}}$$

Find

$$\lim_{n\to\infty} x_n$$

9.1: Introduction to Probability

Definition.
A **sample space** is the set of all possible outcomes of a random process or experiment.
An **event** is a subset of a sample space.
If S is a finite sample space in which all outcomes are equally likely and E is an even in S, then the **probability of** E denoted $P(E)$, is

$$P(E) = \frac{\text{number of outcomes in } E}{\text{total number of outcomes in } S}$$

For any finite set A, $N(A)$ denotes the number of elements in A. Thus,

$$P(E) = \frac{N(E)}{N(S)}$$

Example. Consider a deck of 52 playing cards divided into four suits: ♠ ♥ ♣ ♦

What is the sample space?

What is the event that the chosen card is a red face card?

What is the probability that the chosen card is a red face card?

Example. Consider rolling a pair of 6-sided die.

Write the sample space S of possible outcomes.

Write the event E that the numbers appearing face up have a sum of 6 and find the probability of this event.

The Number of Elements in a List
If m and n are integers and $m \leq n$, then there are $n - m + 1$ integers from m to n inclusive.

Example.

How many three-digit integers are divisible by 5?

What is the probability that a randomly chosen three-digit integer is divisible by 5?

Example. Let $A[1], A[2], \ldots, A[n]$ be a one-dimensional array, where n is a positive integer.

Suppose the array is cut at a middle value $A[m]$ so that two subarrays are formed:

$$A[1], A[2], \ldots, A[m] \text{ and } A[m+1], A[m+2], \ldots, A[n]$$

How many elements does each subarray have?

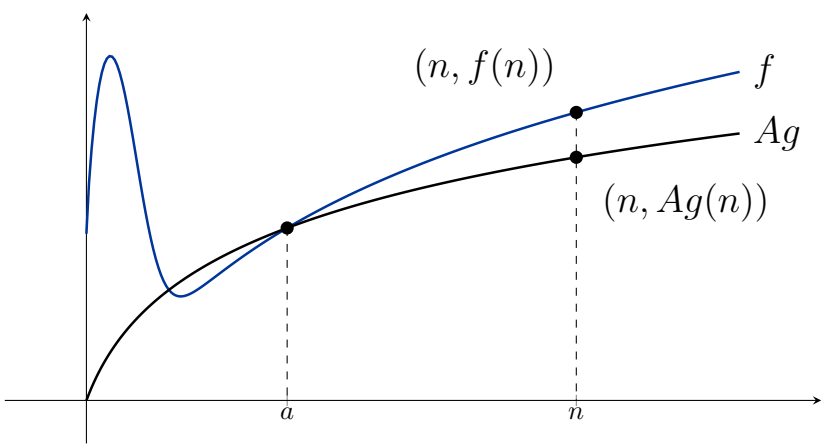
What is the probability that a randomly chosen element of the array has an even subscript?

11.2: Big-$\mathcal{O}$, Big-Omega, and Big-Theta Notations

Definition.
Let f and g be real-valued functions defined on the same set of nonnegative integer, with $g(n) \geq 0$ for every integer $n \geq r$, where r is a positive real number. Then f is of order at least g, written $f(n)$ is $\Omega(g(n))$ if, and only if, there exists positive real numbers A and $a \geq r$ such that

$$Ag(n) \leq f(n) \text{ for every integer } n \geq a.$$

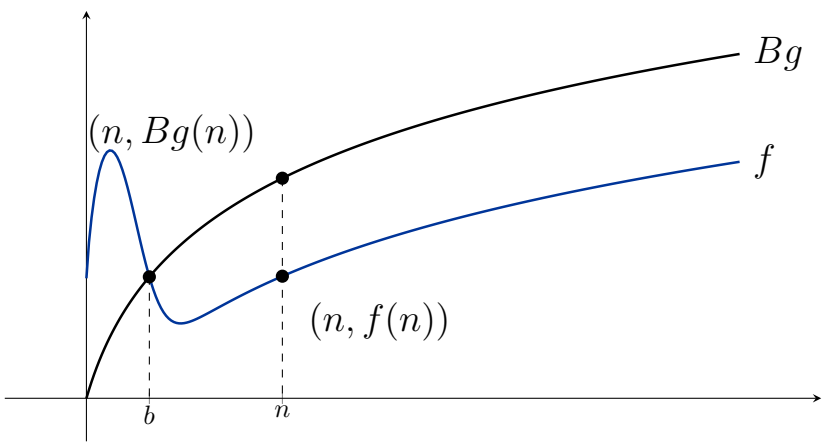
$f(n)$ is $\Omega(g(n))$



Definition.
Let f and g be real-valued functions defined on the same set of nonnegative integer, with $g(n) \geq 0$ for every integer $n \geq r$, where r is a positive real number. Then f is of order at most g, written $f(n)$ is $\mathcal{O}(g(n))$ if, and only if, there exists positive real numbers B and $b \geq r$ such that

$$0 \leq f(n) \leq Bg(n) \text{ for every integer } n \geq b.$$

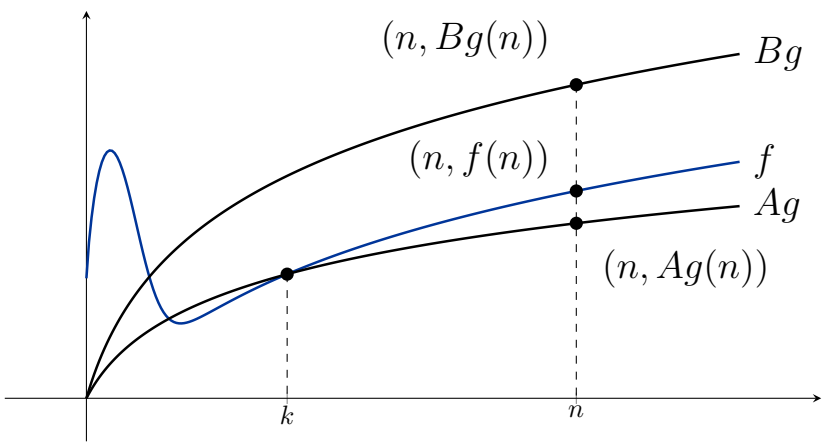
$f(n)$ is $\mathcal{O}(g(n))$



Definition.
Let f and g be real-valued functions defined on the same set of nonnegative integer, with $g(n) \geq 0$ for every integer $n \geq r$, where r is a positive real number. Then f is of order g, written $f(n)$ is $\Theta(g(n))$ if, and only if, there exists positive real numbers A, B and $k \geq r$ such that

$$Ag(n) \leq f(n) \leq Bg(n) \text{ for every integer } n \geq k.$$

$f(n)$ is $\Theta(g(n))$



Example. Use Θ-notation to express the statement

$$4n^6 \leq 17n^6 - 45n^3 + 2n + 8 \leq 30n^6 \text{ for every integer } n \geq 3.$$

Example. Use Ω-notation to express the statement

$$\frac{11}{4}n^2 \leq 3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \text{ for every integer } n \geq 2$$

Use $\mathcal{O}$-notation to express the statement

$$0 \leq 3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2 \leq 6n^2 \text{ for every integer } n \geq 1.$$

Justify the statement: $3\left(\left\lfloor \frac{n}{4} \right\rfloor\right)^2 + 5n^2$, is $\Omega\big(n^2\big)$.

10.1: Trails, Paths, and Circuits

Definition.

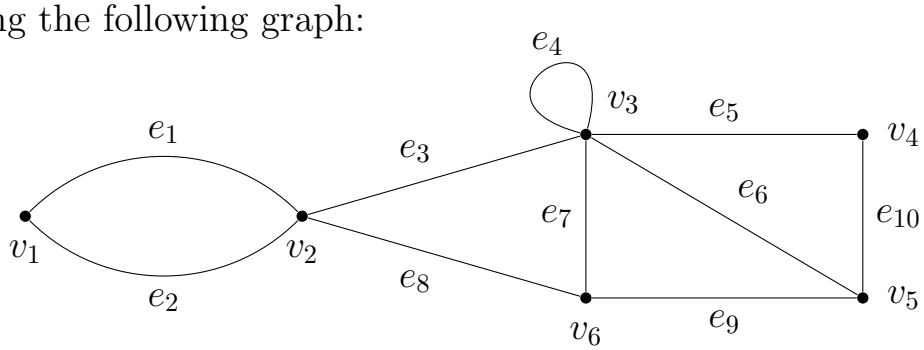
- Let G be a graph, and let v and w be vertices in G.
- A **walk from** v **to** w is a finite alternating sequence of adjacent vertices and edges of G. A walk has the form

$$v_0\ e_1\ v_1\ e_2\ \ldots\ v_{n-1}\ e_n\ v_n,$$

  where the v's represent vertices, the e's represent edges, $v_0 = v$, $v_n = w$, and for each $i = 1, 2, \ldots, n$, v_{i-1} and v_i are the endpoints of e_i. The **trivial walk from** v **to** v consists of the single vertex v.
- A **trail from** v **to** w is a walk from v to w that does not contain a repeated edge.
- A **path from** v **to** w is a trail that does not contain a repeated vertex.
- A **closed walk** is a walk that starts and ends at the same vertex.
- A **circuit** is a closed walk that contains at least one edge and does not contain a repeated edge.
- A **simple circuit** is a circuit that does not have any other repeated vertex except the first and last.

	Repeated Edge?	Repeated Vertex?	Starts and Ends at the Same Point?	Must Contain at Least One Edge?
Walk	allowed	allowed	allowed	no
Trail	no	allowed	allowed	no
Path	no	no	no	no
Closed walk	allowed	allowed	yes	no
Circuit	no	allowed	yes	yes
Simple circuit	no	first and last only	yes	yes

Example. Using the following graph:



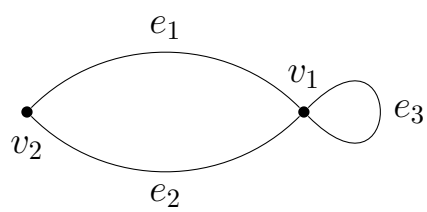
determine which of the following walks are trails, paths, circuits, or simple circuits.

$v_1\, e_1\, v_2\, e_3\, e_4\, v_3\, e_5\, v_4$ $\qquad$ $e_1\, e_3\, e_5\, e_5\, e_6$ $\qquad$ $v_2\, v_3\, v_4\, v_5\, v_3\, v_6\, v_2$

$v_2\, v_3\, v_4\, v_5\, v_6\, v_2$ $\qquad$ $v_1\, e_1\, v_2\, e_1\, v_1$ $\qquad$ v_1

Definition.
A graph H is said to be a **subgraph** of a graph G if, and only if, every vertex in H is also a vertex in G, every edge in H is also an edge in G, and every edge in H has the same endpoints as it has in G.

Example. Based on the following graph:



How many paths/trails/walks are there from v_1 to v_2?

List all subgraphs.

Definition.
Let G be a graph. Two **vertices v and w of G are connected** if, and only if, there is a walk from v to w. The **graph G is connected** if, and only if, given *any* two vertices v and w in G, there is a walk from v to w.

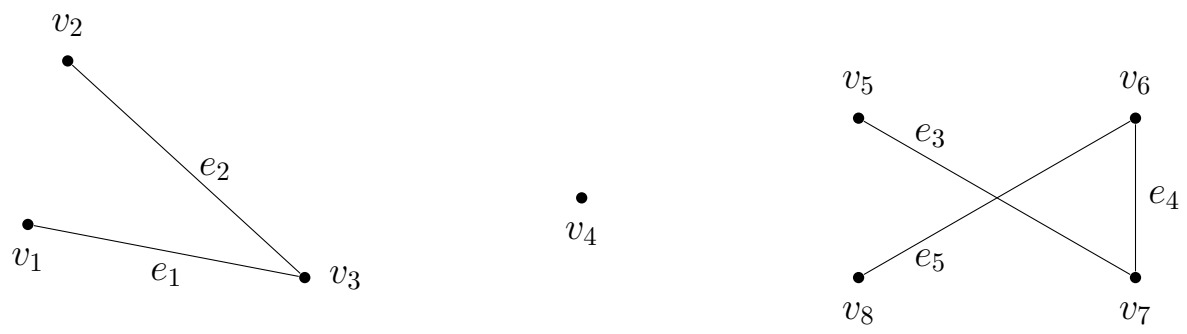
Lemma 10.1.1
Let G be a graph.

1. If G is connected, then any two distinct vertices of G can be connected by a path.
2. If vertices v and w are part of a circuit in G and one edge is removed from the circuit, then there still exists a trail from v to w in G.
3. If G is connected and G contains a circuit, then an edge of the circuit can be removed without disconnecting G.

Definition.
A graph H is a **connected component** of a graph G if, and only if,

1. H is a subgraph of G;
2. H is connected; and
3. no connected subgraph of G has H as a subgraph and contains vertices or edges that are not in H.

Example. Find all connected components of the following graph G



Definition.
Let G be a graph. An **Euler circuit** for G is a circuit that contains every vertex and every edge of G. That is, an Euler circuit for G is a sequence of adjacent vertices and edges in G that has at least one edge, starts and ends at the same vertex, uses every vertex of G at least once, and uses every edge of G exactly once.

Theorem 10.1.2
If a graph has an Euler circuit, then every vertex of the graph has positive even degree.

Theorem 10.1.3
If a graph G is connected and the degree of every vertex of G is a positive even integer, then G has an Euler circuit.

Theorem 10.1.4
A graph G has an Euler circuit if, and only if, G is connected and every vertex of G has positive even degree.

Definition.
Let G be a graph, and let v and w be two distinct vertices of G. An **Euler trail from v to w** is a sequence of adjacent edges and vertices that starts at v, ends at w, passes through every vertex of G at least once, and traverses every edge of G exactly once.

Definition.
Given a graph G, a **Hamiltonian circuit** for G is a simple circuit that includes every vertex of G. That is, a Hamiltonian circuit for G is a sequence of adjacent vertices and distinct edges in which every vertex of G appears exactly once, except for the first and the last, which are the same.

Proposition 10.1.6
If a graph G has a Hamiltonian circuit, then G has a subgraph H with the following properties:

- H contains every vertex of G.
- H is connected.
- H has the same number of edges as vertices.
- Every vertex of H has degree 2.

Definition.
If G is a simple graph, the **complement of** G, denoted G', is obtained as follows: The vertex set of G' is identical to the vertex set of G. However, two distinct vertices v and w of G' are connected by an edge if, and only if, v and w are not connected by an edge in G.

10.3: Isomorphisms of Graphs

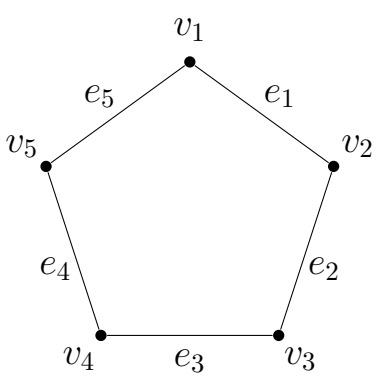
Definition.
Let G and G' be graphs with vertex sets $V(G)$ and $V(G')$ and edge sets $E(G)$ and $E(G')$, respectively. G **is isomorphic to** G' if, and only if, there exist one-to-one correspondences $g : V(G) \to V(G')$ and $h : E(G) \to E(G')$ that preserve the edge-endpoint functions of G and G' in the sense that for each $v \in V(G)$ and $e \in E(G)$,

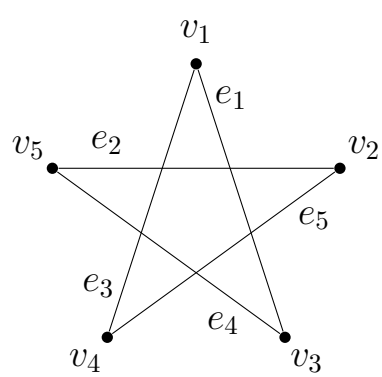
$$v \text{ is an endpoint of } e \Leftrightarrow g(v) \text{ is an endpoint of } h(e).$$

If G and G' are simple (no loops or parallel edges), then G is isomorphic to G' if, and only if, there exists $g : V(G) \to V(G')$ such that

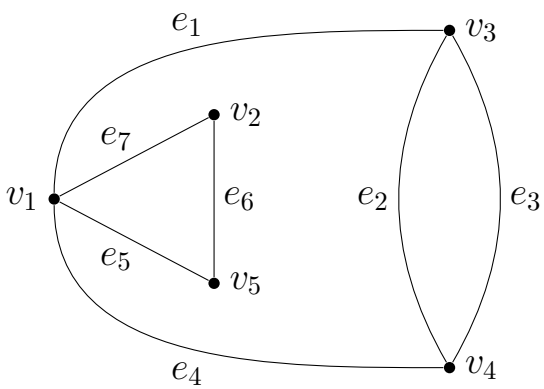
$$\{u, v\} \text{ is an edge in } G \Leftrightarrow \{g(u), g(v)\} \text{ is an edge in } G'.$$

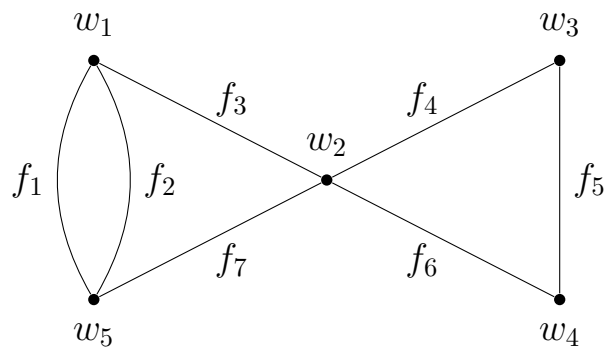
Example. Show that the following graphs are isomorphic:





Example. Show that the following graphs are isomorphic:

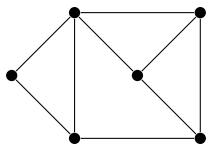
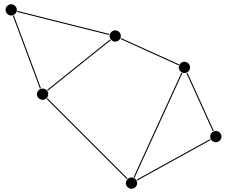
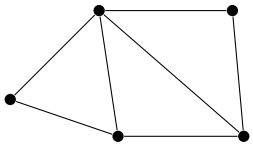
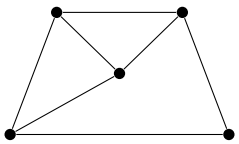
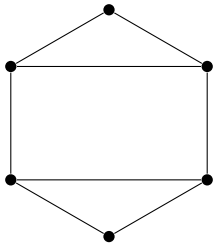
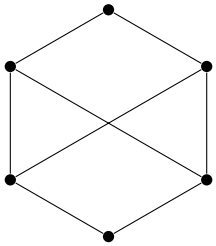




Theorem 10.3.1
Let S be a set of graphs and let R be the relation of graph isomorphisms on S. Then R is an equivalence relation on S.

Definition.
A property P is called an **invariant for graph isomorphism** if, and only if, given any graphs G and G', if G has property P and G' is isomorphic to G, then G' has property P.

Theorem 10.3.2
Each of the following properties is an invariant for graph isomorphism, where n, m and k are all nonnegative integer:

1. has n vertices
2. has m edges
3. has a vertex of degree k
4. has m vertices of degree k
5. has a circuit of length k
6. has a simple circuit of length k
7. has m simple circuits of length k
8. is connected
9. has an Euler circuit
10. has a Hamiltonian circuit.

Example. For the following pairs of graphs, determine if the graphs are isomorphic. Explain why or why not.